

Recap

K means $\rightarrow$ hard clustering $\rightarrow$ 1 datapoint to 1 cluster
 GMM $\rightarrow$ soft clustering $\rightarrow$ EM



$$p(x|\theta) = \pi_0 N_0 + \pi_1 N_1 + \dots + \pi_K N_K$$

Step 0: initialize μ, Σ, π

$$z \sim p(z)$$

$$x \sim p(x|z)$$

GMM for clustering
 $p(z_k|x) \leftarrow \text{resp.}$

- Unsupervised methods covered so far:

$$p(x, z) = \underbrace{p(x|z)}_N \cdot \underbrace{p(z)}_{\pi}$$

- Generative vs Discriminative models

logistic regression
 Neural networks

$$p(z_k|x)$$

- Common problem between unsupervised methods covered so far

Both GMM & kmeans $\rightarrow$ ~~k~~

Hierarchical Clustering

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Outline

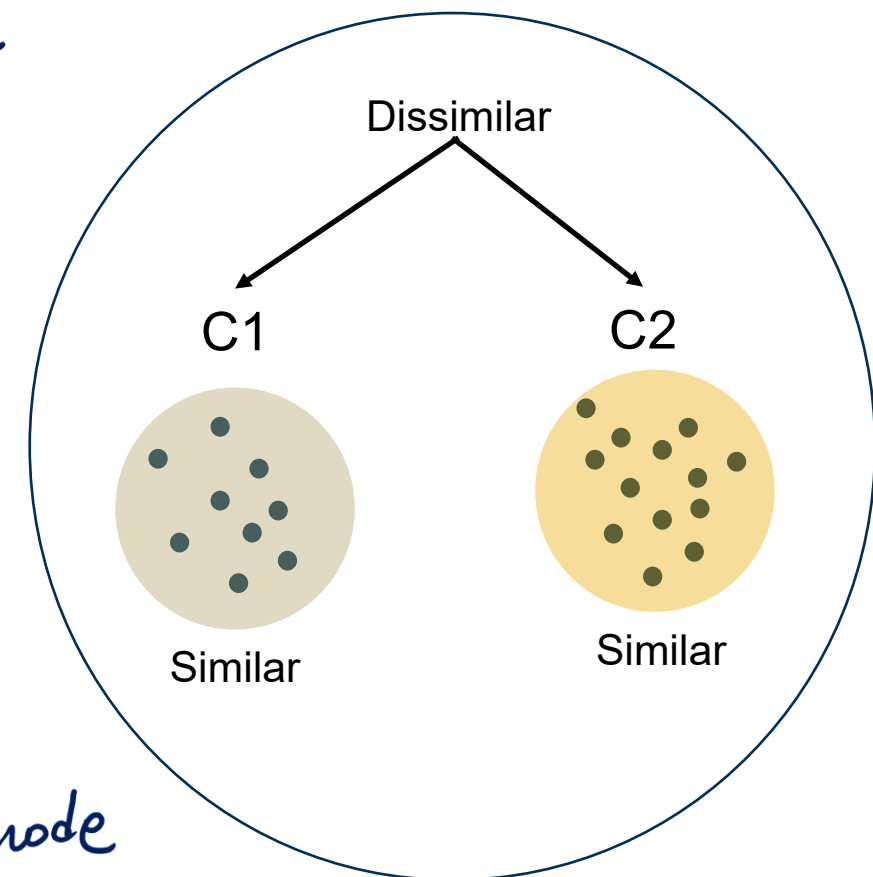
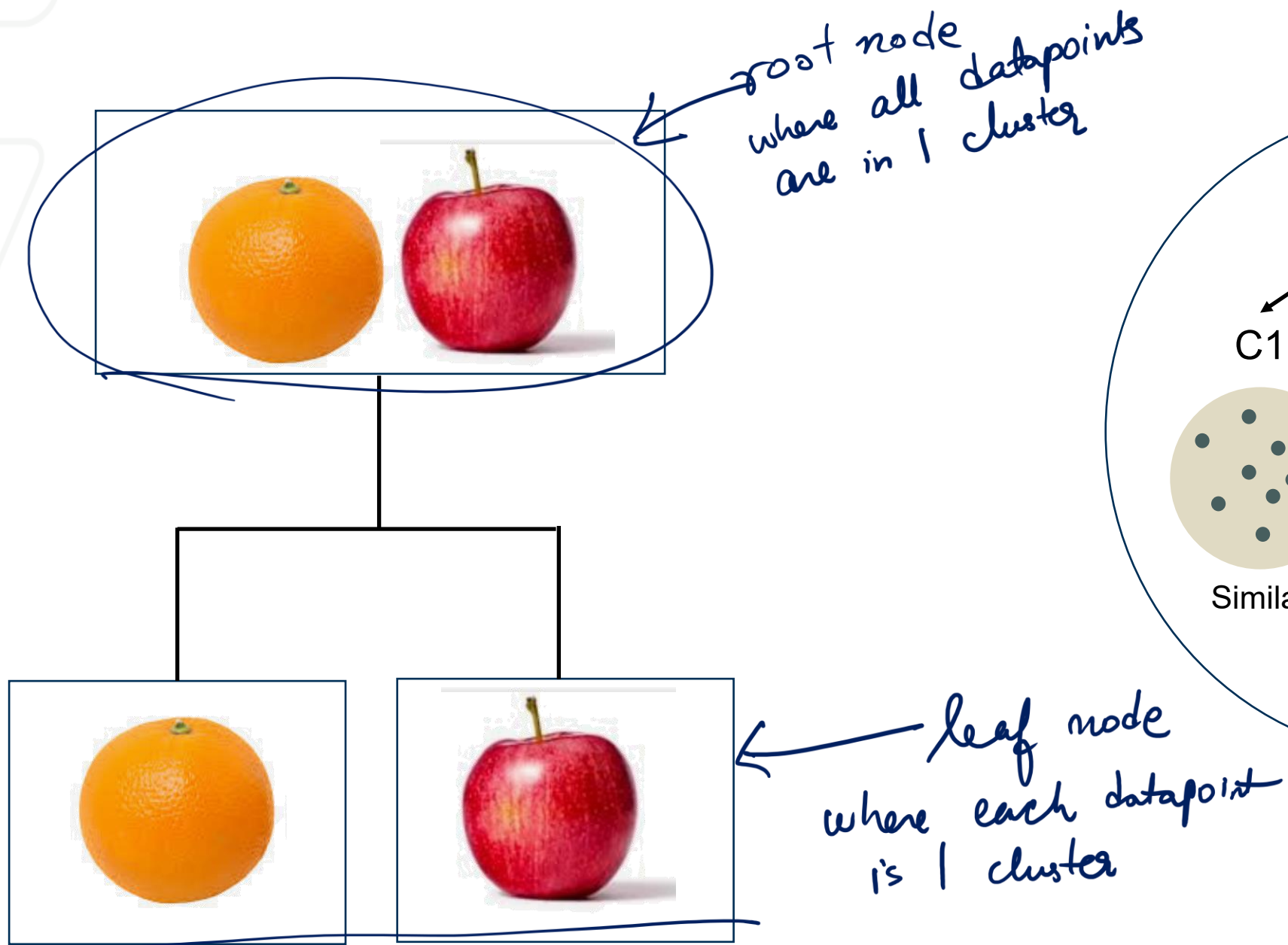
- Overview 
- Bottom-Up vs Top-Down Clustering
- Measuring Distance between Clusters

Hierarchical Clustering vs Partitional Clustering

Agglomerative

Divisive

K-Means



Tree structure (parent-child relationship)

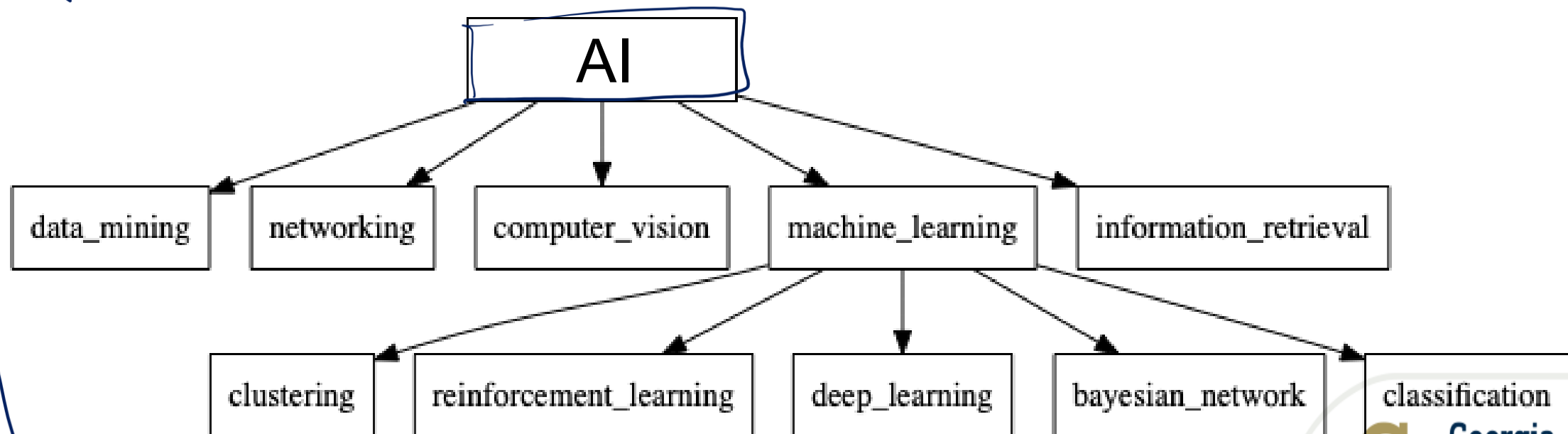
Hierarchical Clustering

- How to organize a set of CS papers into a hierarchy?

Drawback:
Generally works well
for small datasets



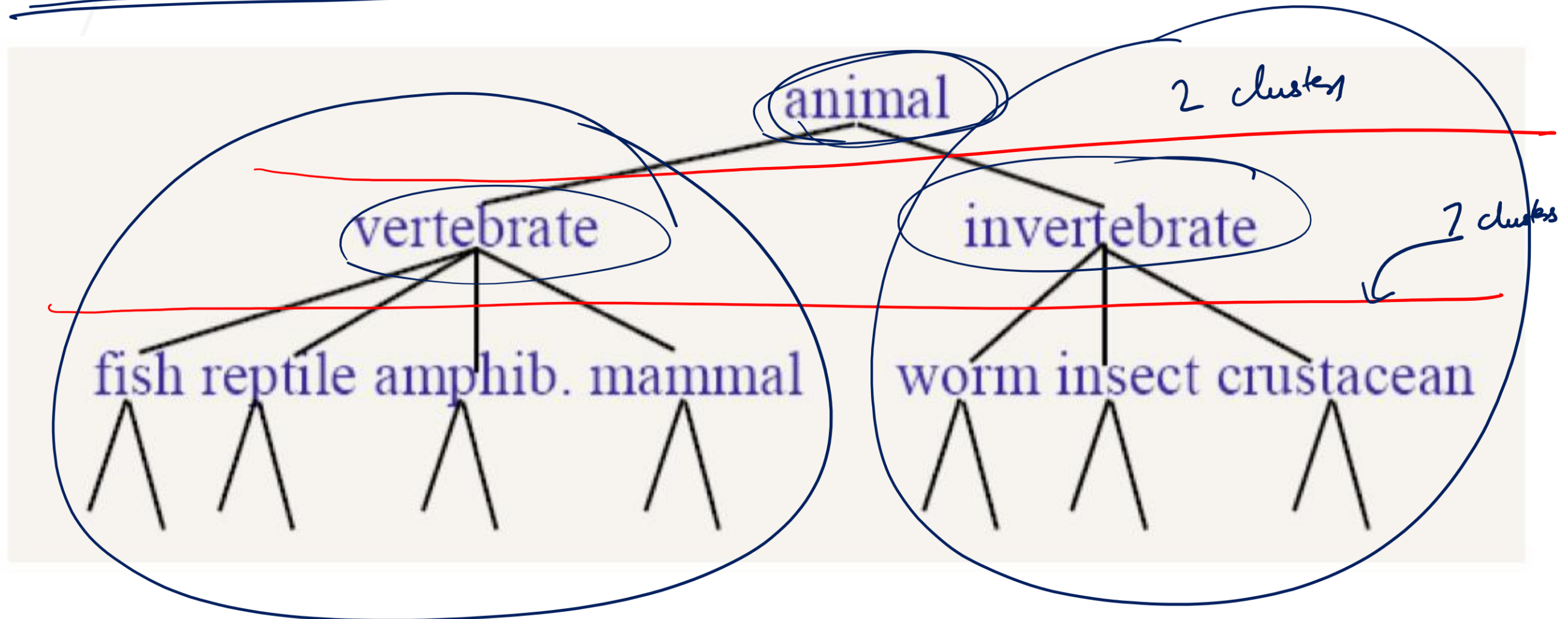
Tree like
structure



Hierarchical Clustering

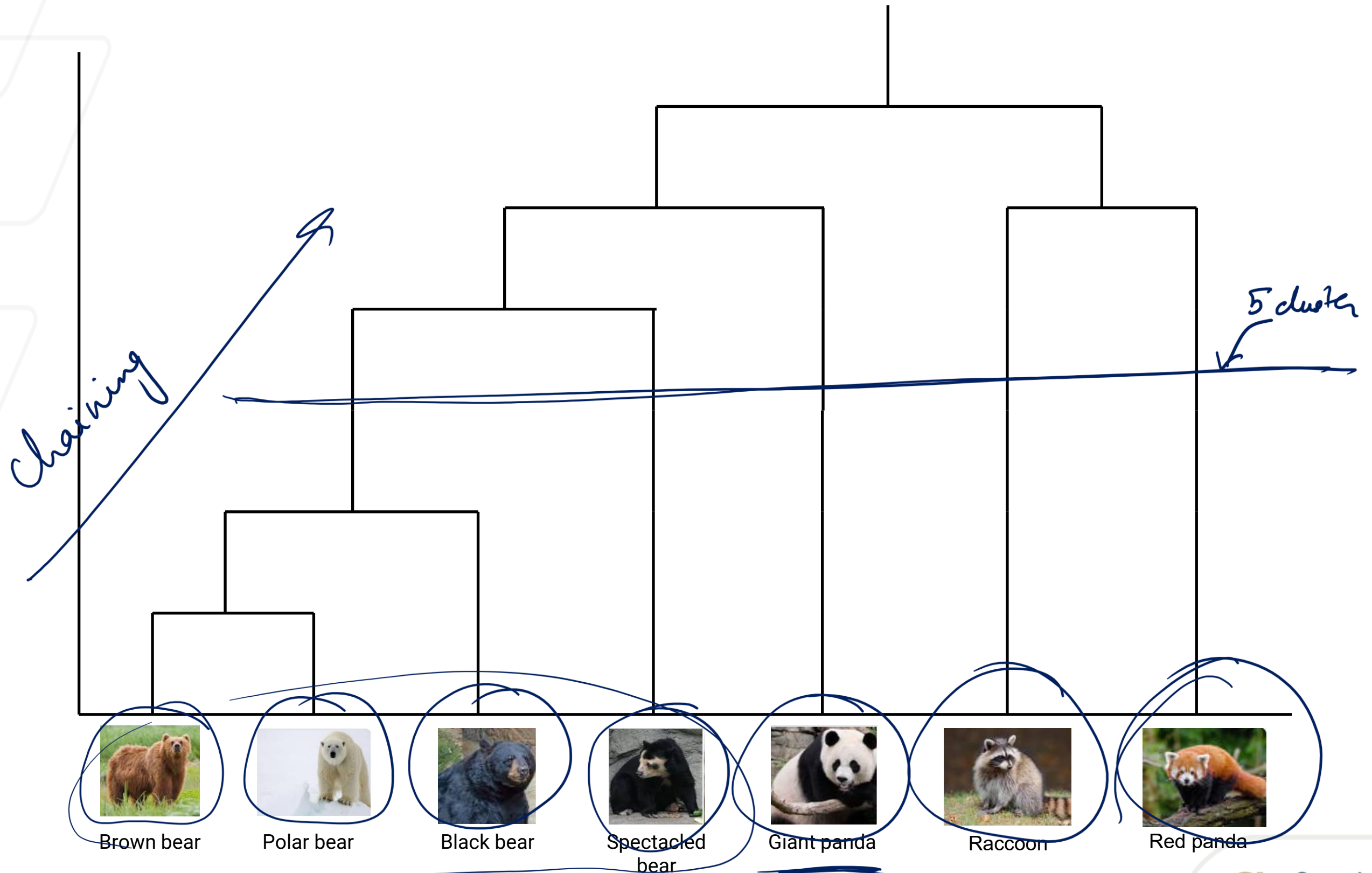
Vertical dendrogram →
make a horizontal
cut → how many
clusters.

- Organize objects into a tree-based hierarchical taxonomy (dendrogram)



- Many applications in the real world
 - Web pages
 - News articles
 - Scientific papers

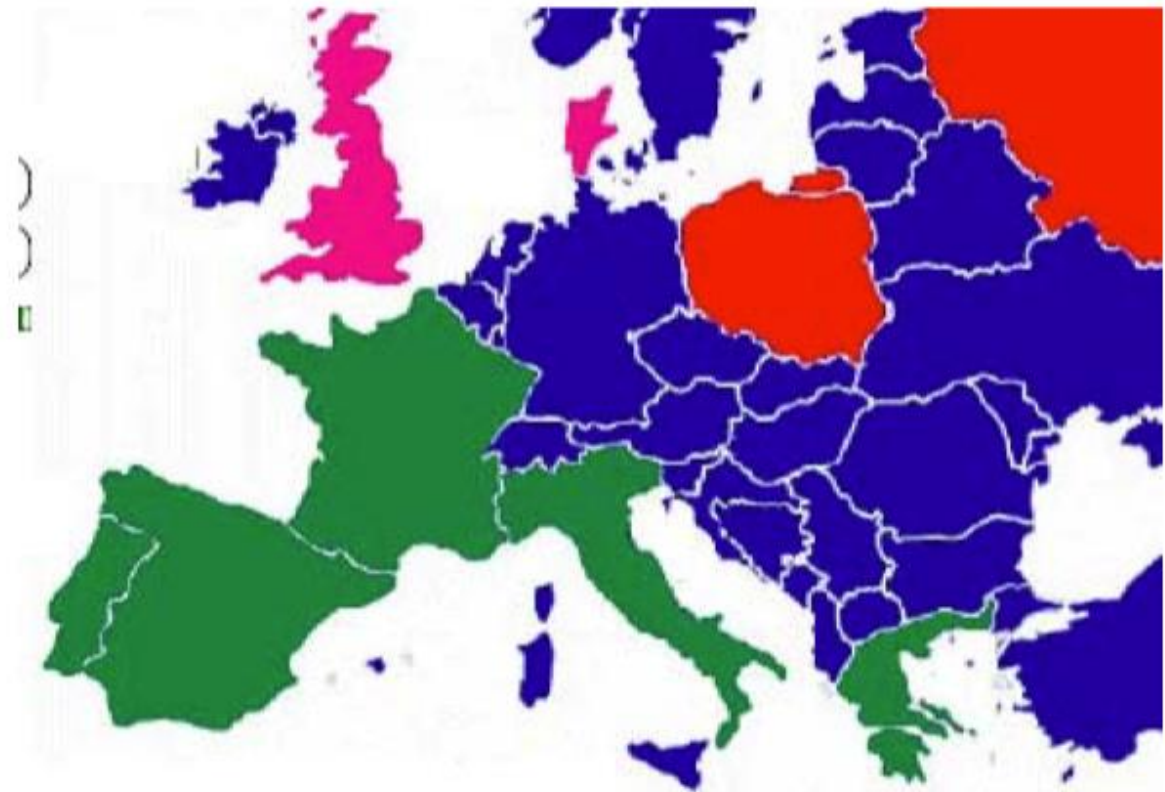
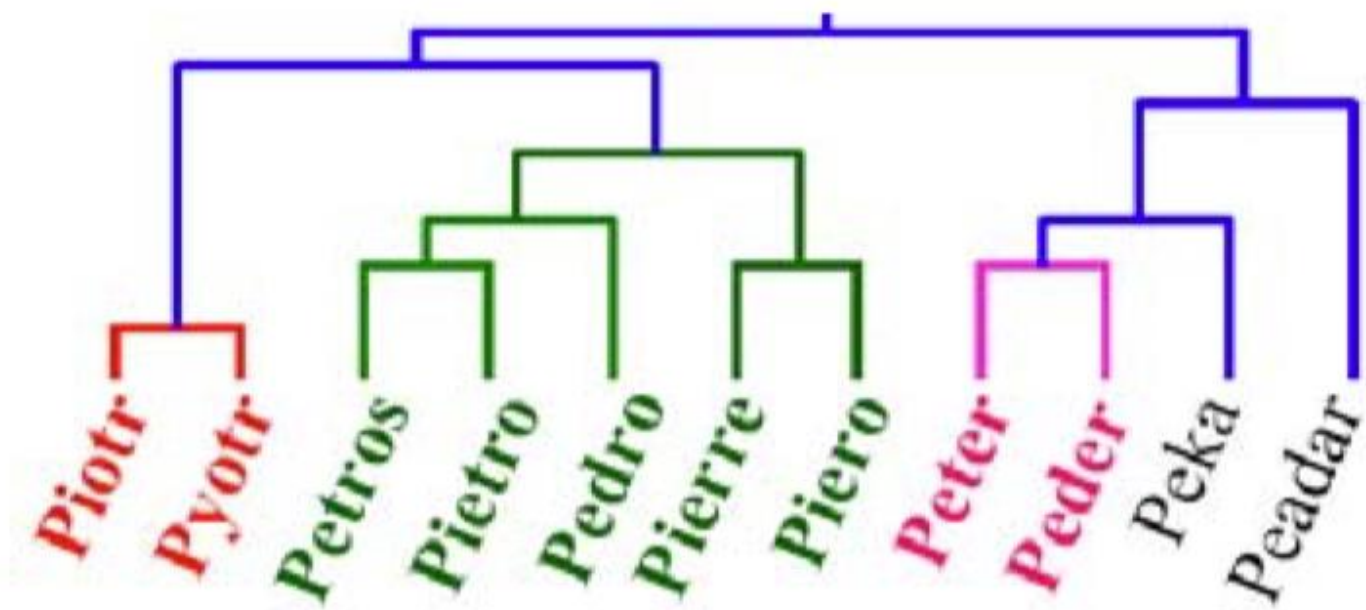
DNA sequencing and hierarchical clustering to find the phylogenetic tree of animal evolution



Using Hierarchical clustering, the researchers were able to place the giant pandas closer to bears

Hierarchical Clustering

- Organizing data at multiple granularities
- Cutting the dendrogram at a desired level leads to a sub-cluster: each connected component forms a cluster



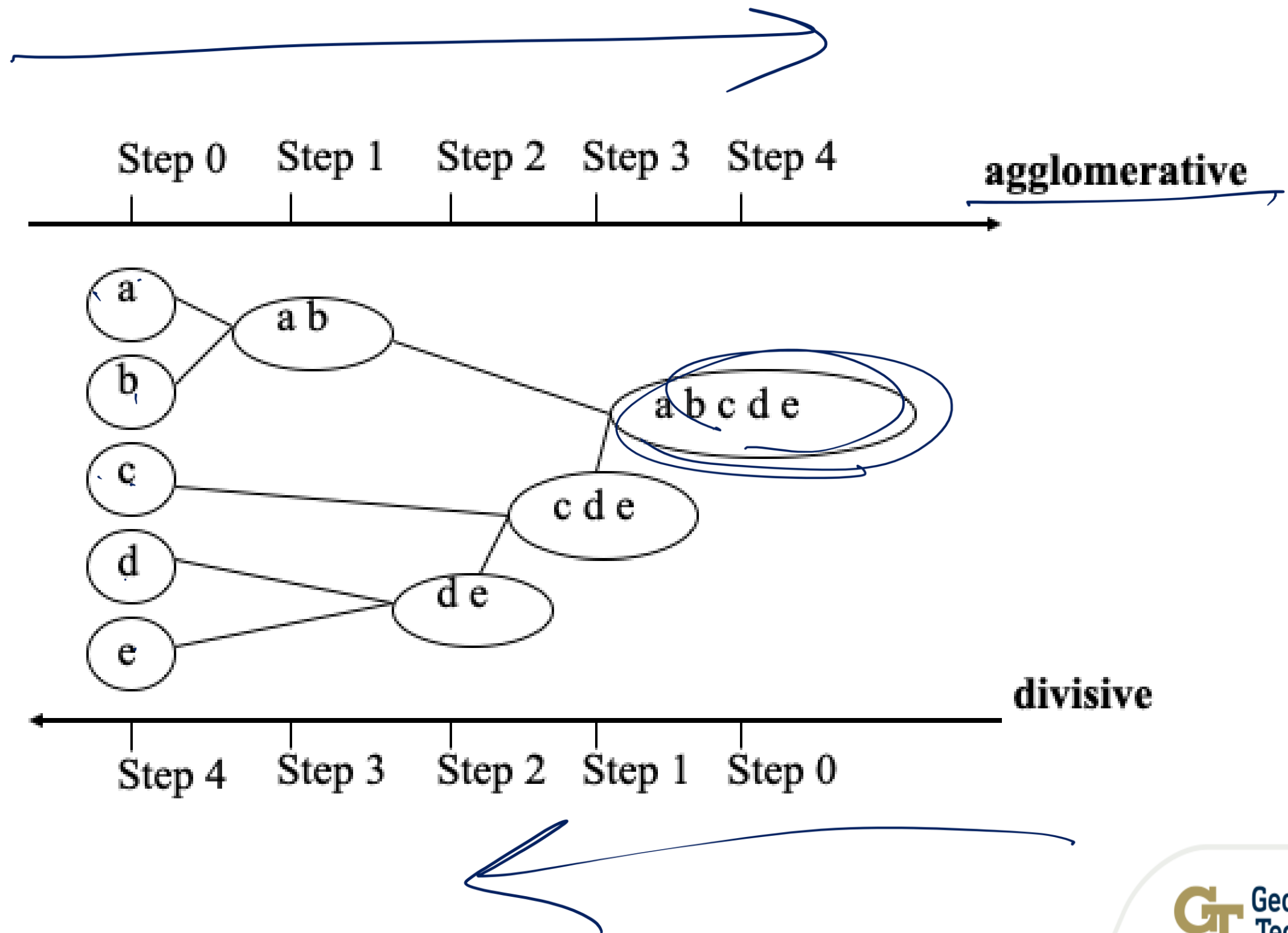
Outline

- Overview
- Bottom-Up vs Top-Down Clustering ←
- Measuring Distance between Clusters

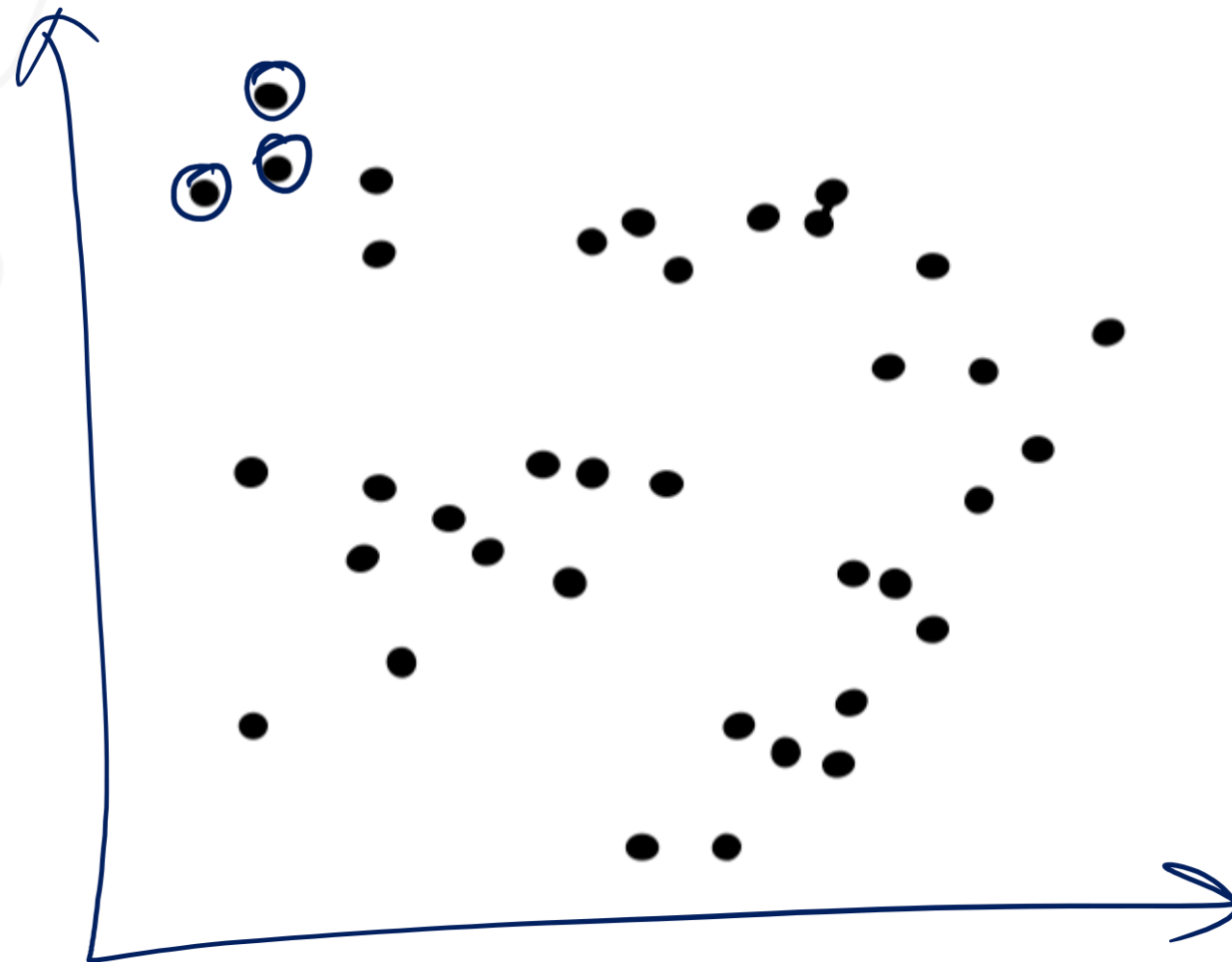
Two Paradigms for Hierarchical Clustering

- Bottom-up Agglomerative Clustering
 - Start by considering each object as a separate cluster
 - Repeatedly join the closest pair of clusters
 - Stop when there is only one cluster left
- Top-Down Divisive Clustering
 - Start by considering all objects as one large cluster
 - Recursively divide each cluster into two sub-clusters
 - Stop when each cluster contains only one object

Bottom-Up v.s. Top-Down

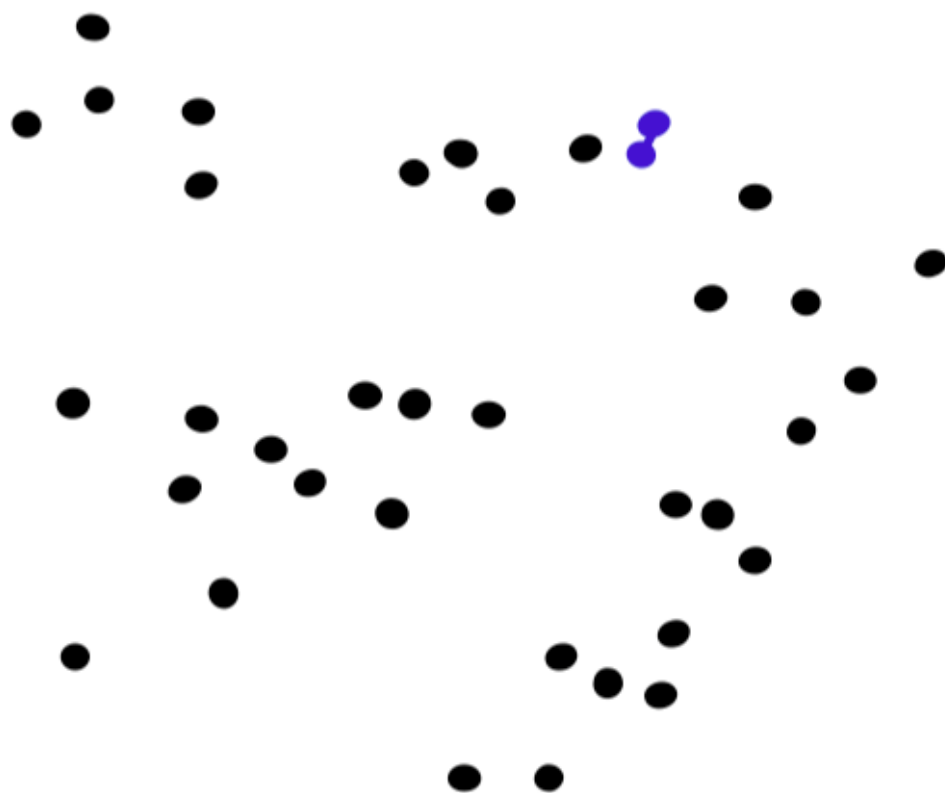


Bottom-Up Agglomerative Clustering

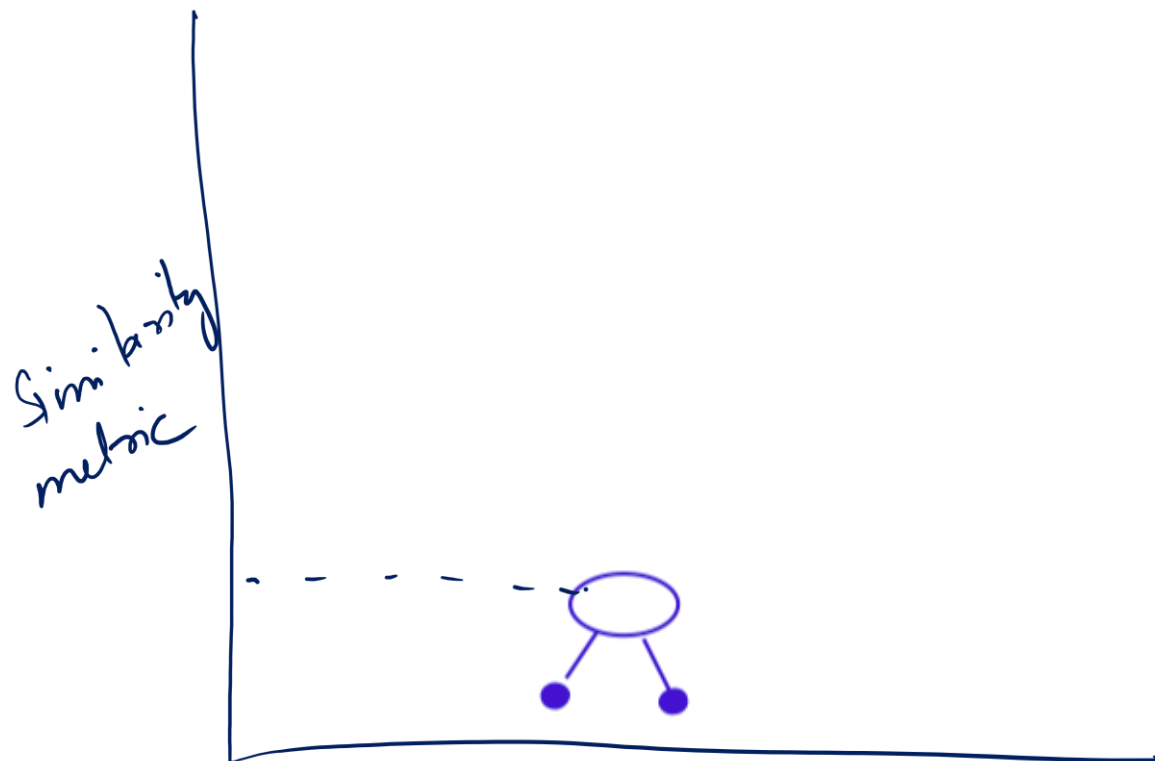


1. Say "Every point is it's own cluster"

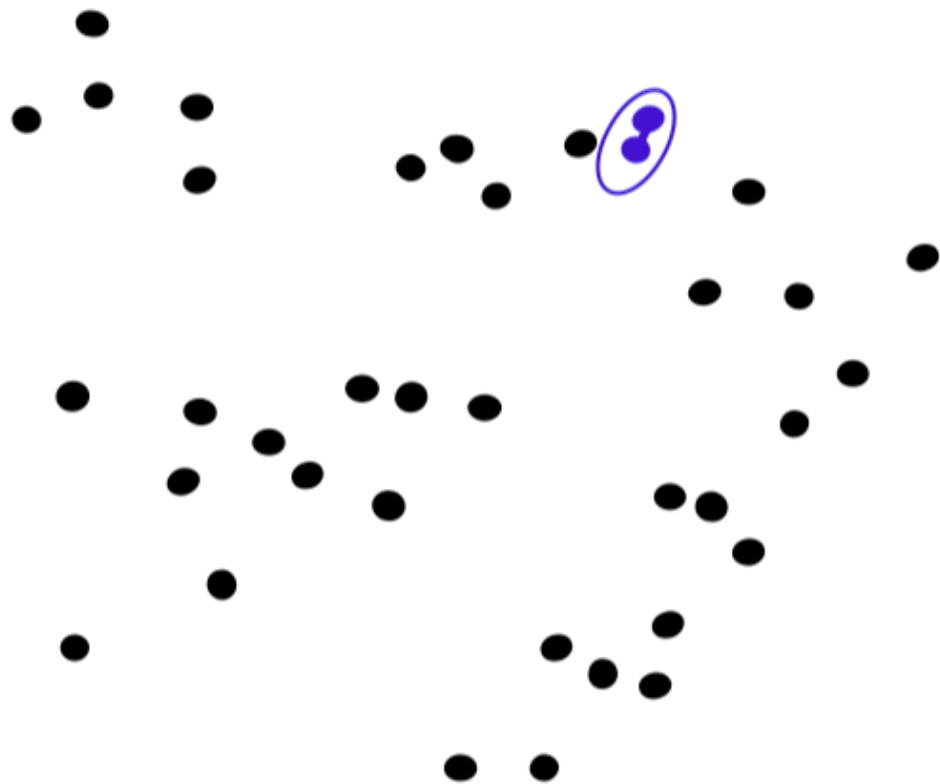
Bottom-Up Agglomerative Clustering



1. Say "Every point is it's own cluster"
2. Find "most similar" pair of clusters



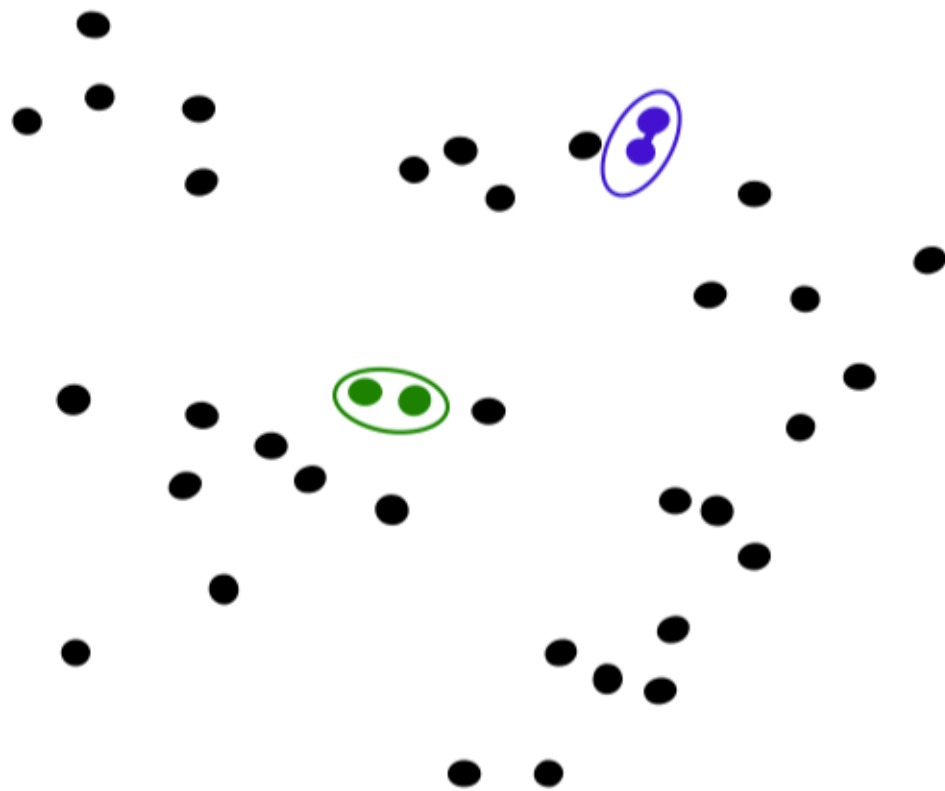
Bottom-Up Agglomerative Clustering



1. Say "Every point is it's own cluster"
2. Find "most similar" pair of clusters
3. Merge it into a parent cluster



Bottom-Up Agglomerative Clustering



1. Say "Every point is it's own cluster"
2. Find "most similar" pair of clusters
3. Merge it into a parent cluster
4. Repeat



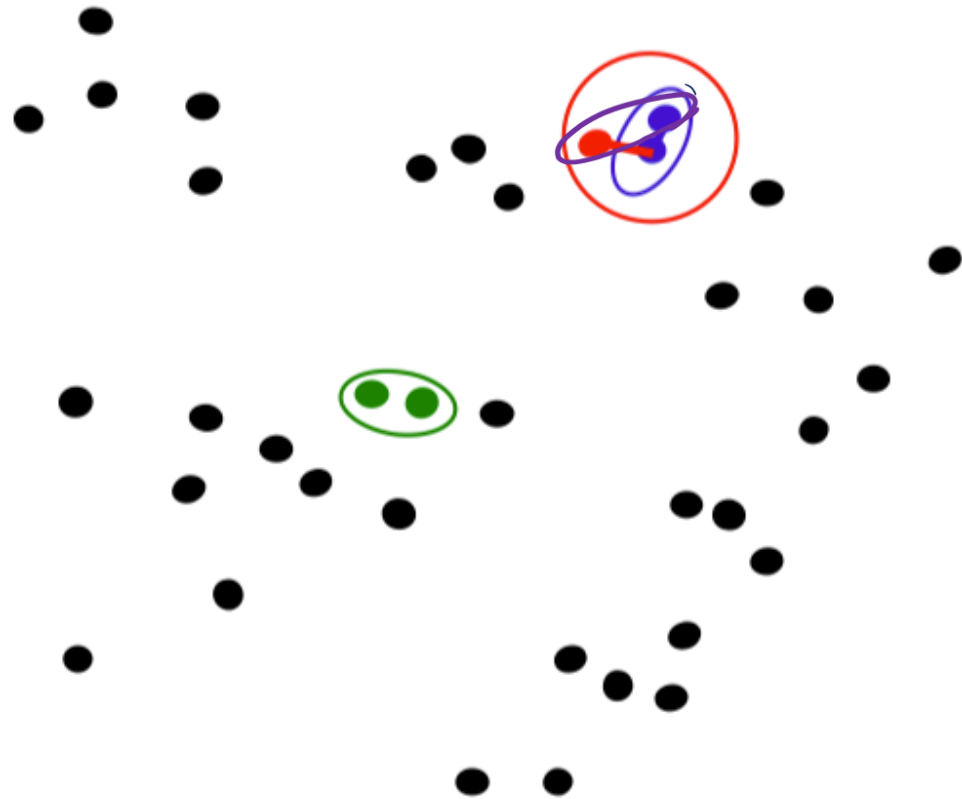
Bottom-Up Agglomerative Clustering

average linkage

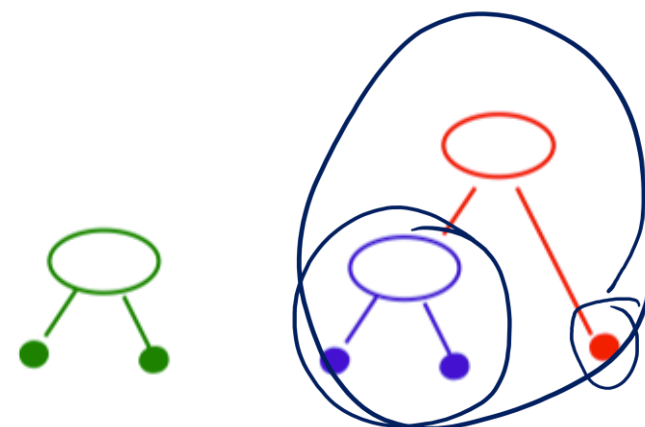
single linkage

complete linkage

centroid linkage



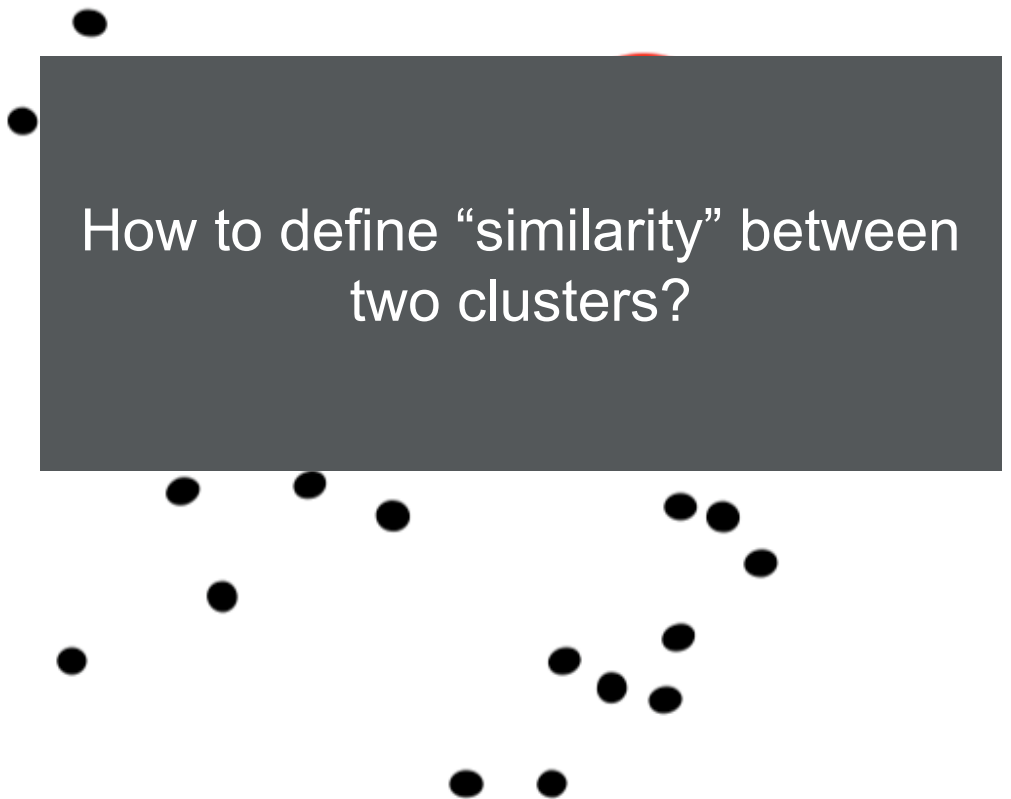
1. Say "Every point is it's own cluster"
2. Find "most similar" pair of clusters
3. Merge it into a parent cluster
4. Repeat



Outline

- Overview
- Bottom-Up vs Top-Down Clustering
- Measuring Distance between Clusters ←

Key Question: Similarity Function

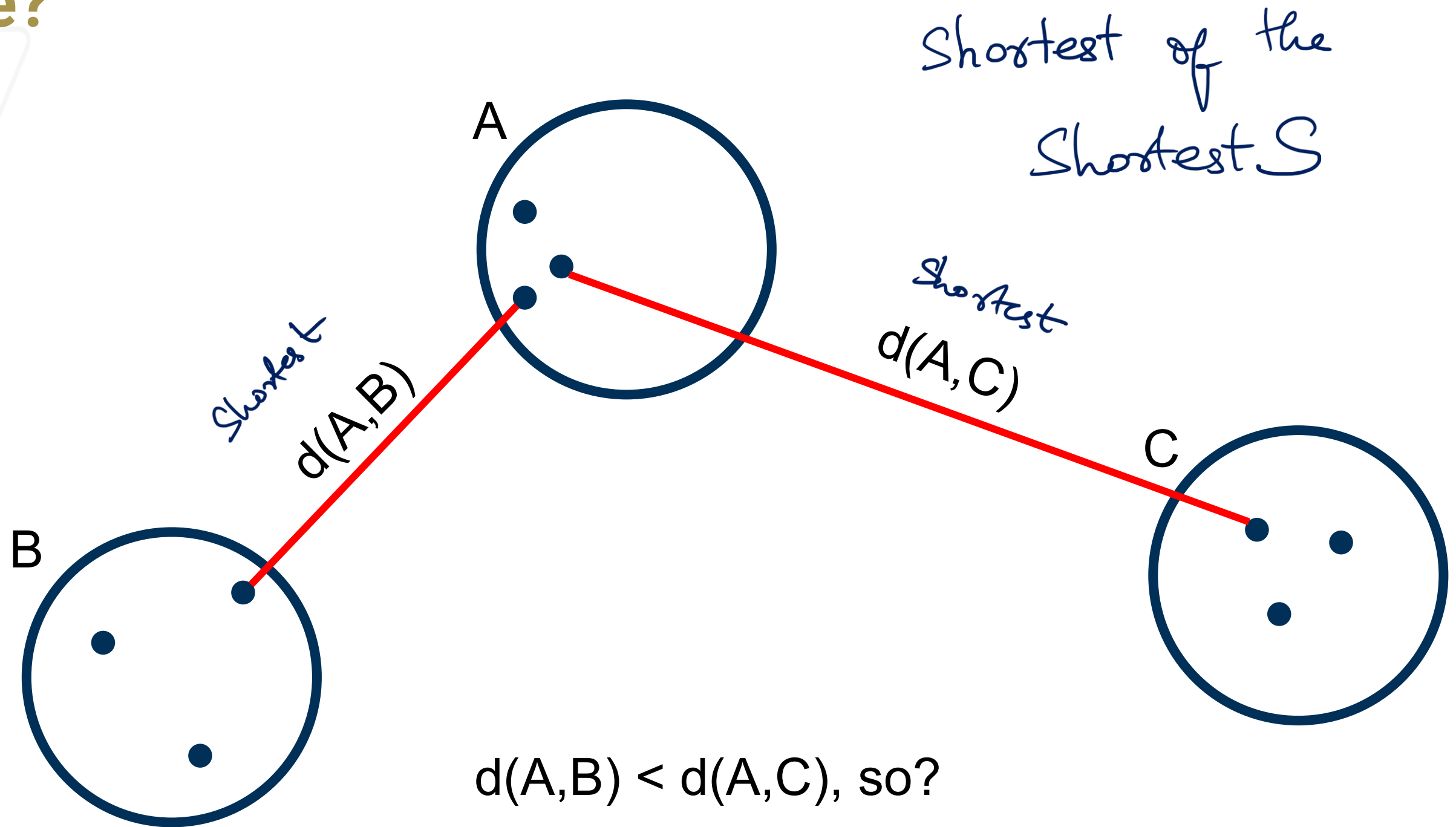


How to define “similarity” between two clusters?

1. Say “Every point is it’s own cluster”
2. Find “most similar” pair of clusters
3. Merge it into a parent cluster
4. Repeat

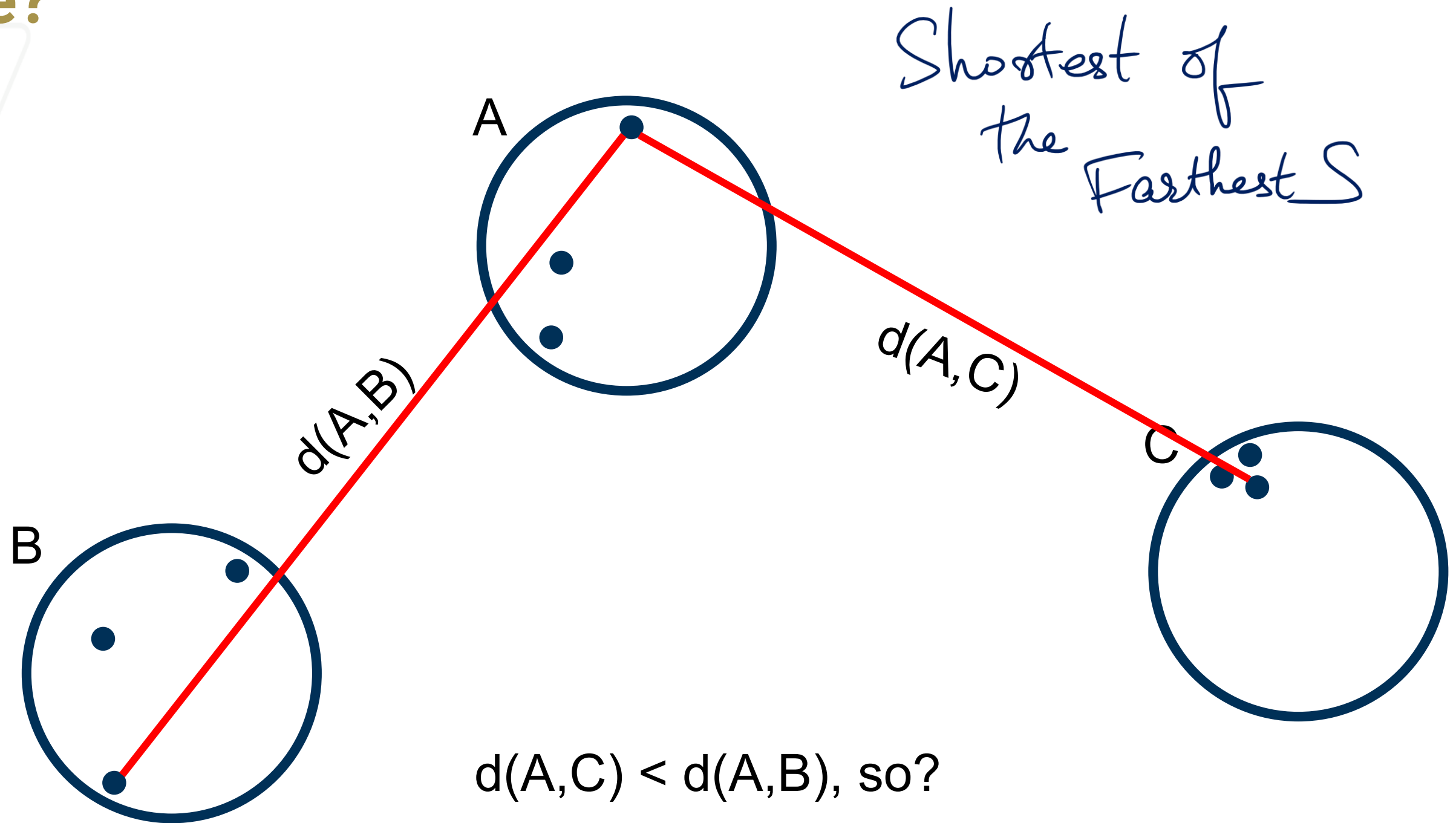


I am going to merge A with either B or C. Which one?



Single Link

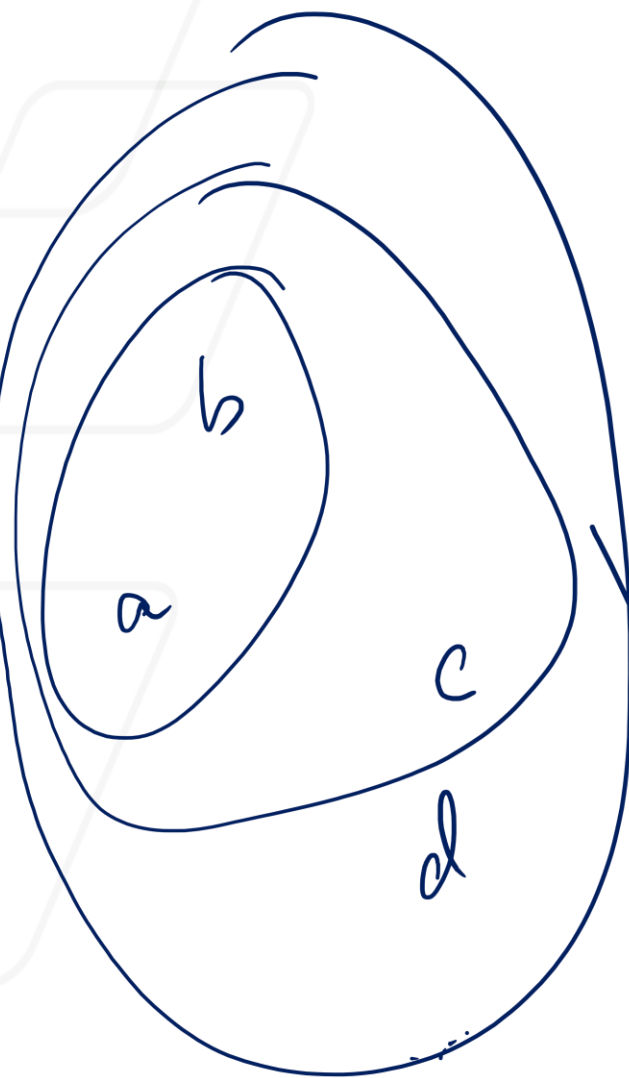
I am going to merge A with either B or C. Which one?



Complete Link

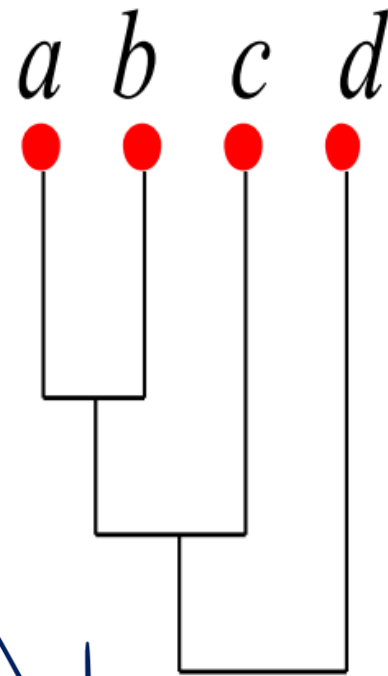
- **Single link:** A chain of points can be extended for long distances without regard to the overall shape of the emerging cluster. This effect is called *chaining*. It is also sensitive to outliers. It is faster in general.
 - Best when **detecting elongated, irregular shapes**. Example: You're clustering cities along a river. Single-link will connect them in order, capturing the *path-like structure*.
 - **When clusters are non-spherical and winding** (like spirals, curves).
- **Complete link:** Clusters are split into two groups of roughly equal size when we cut the dendrogram at the last merge. In general, this is a more useful organization of the data than a clustering with chains. It avoids chaining and more robust to outliers. Generally slower.
 - Best when **you want compact, well-separated groups**. Example: Customer segmentation in marketing — you want customers in the same group to be similar in *all* features, not just one.
 - **Outlier robustness:** Because it considers the farthest points, clusters won't "stretch" easily.
- **Average link:** When you don't know which one may be better for you, start it with the average link method.

Dendrograms

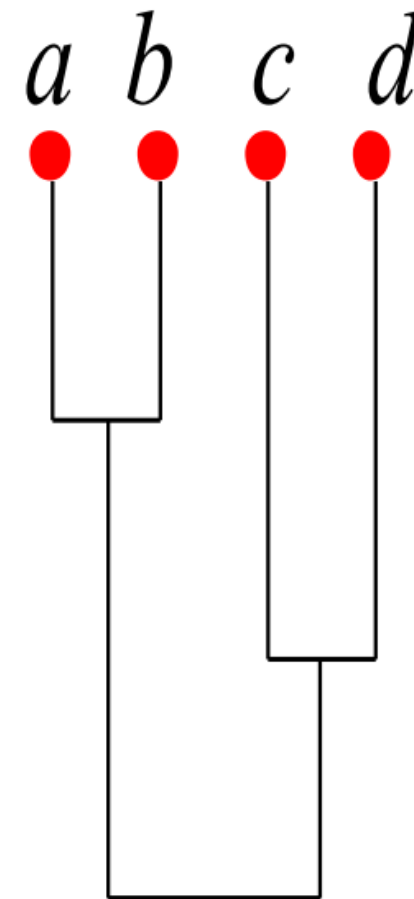


*outliers
& chaining*

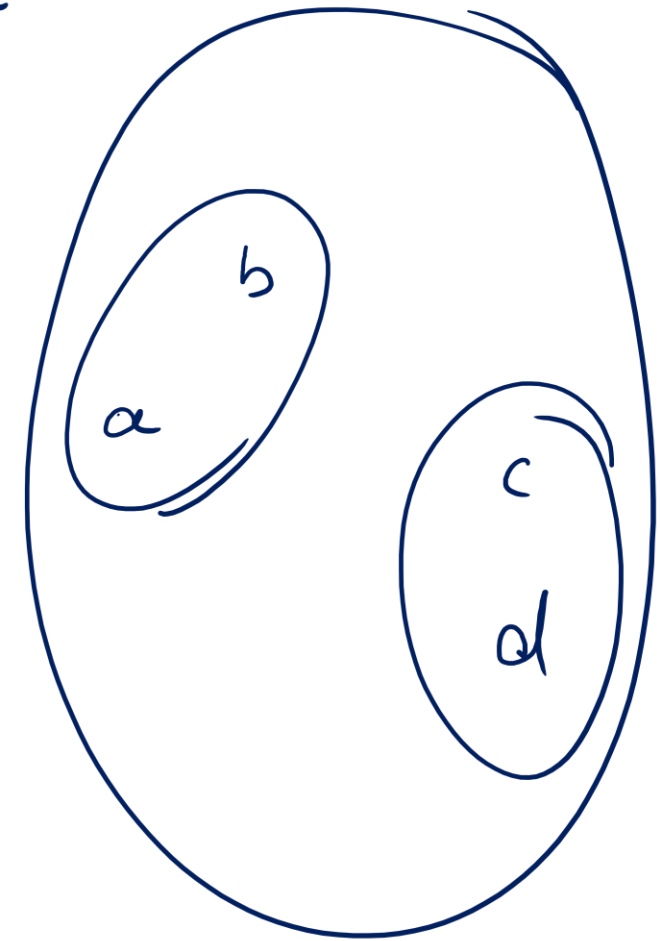
Single-Link



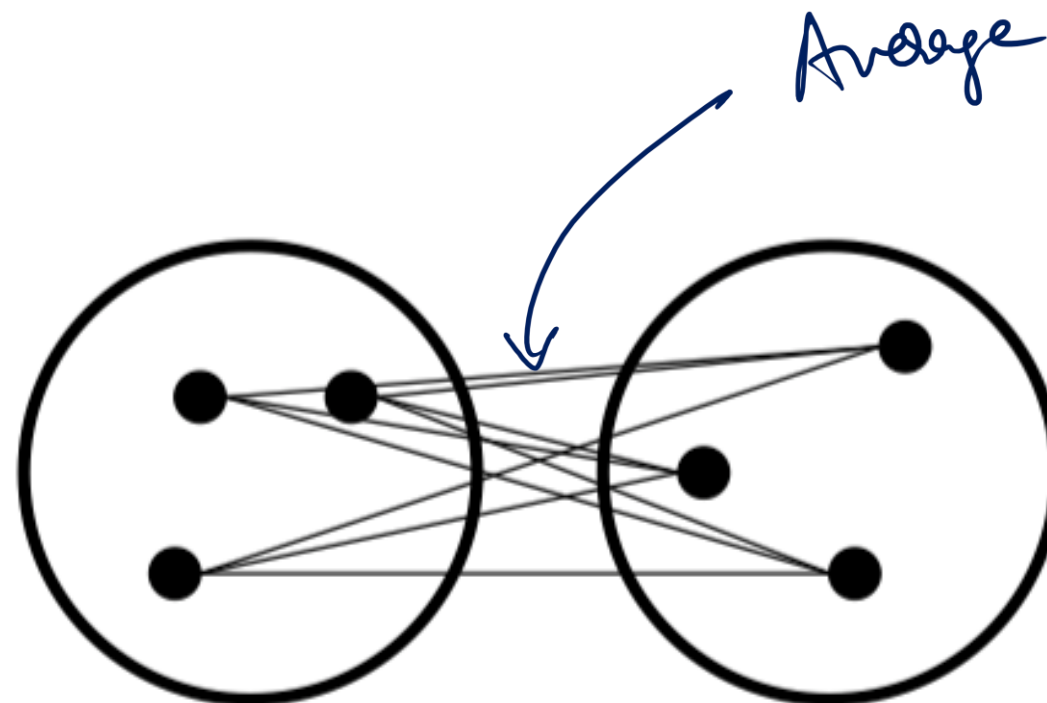
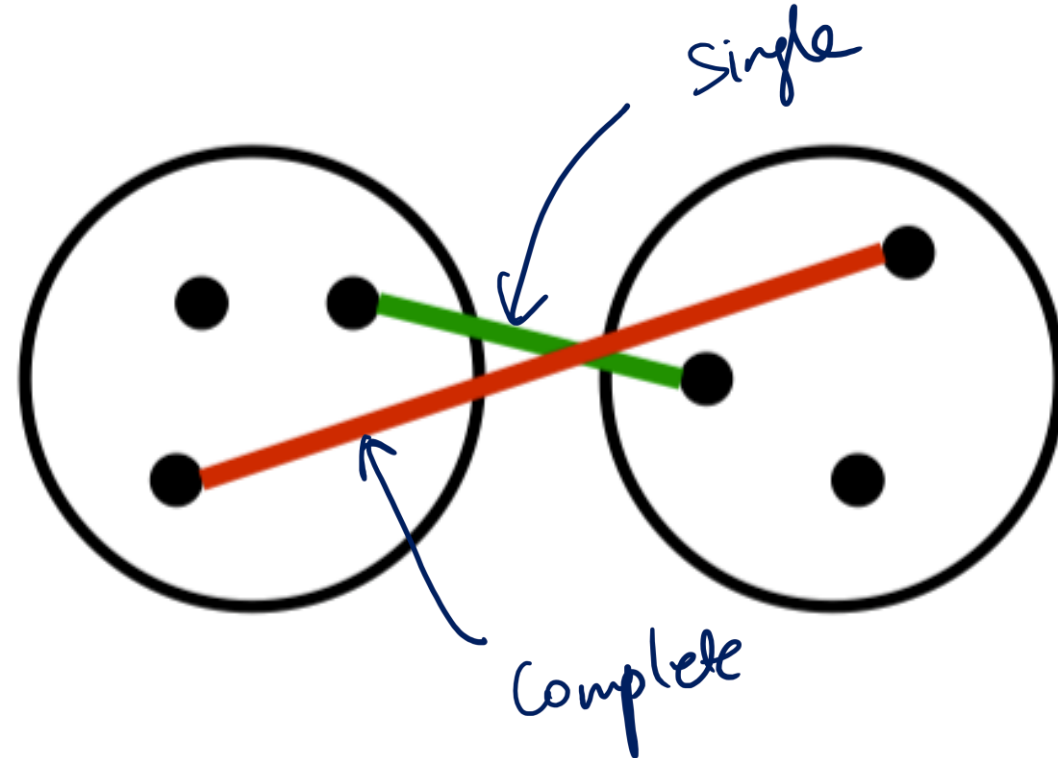
Complete-Link



X → more balanced



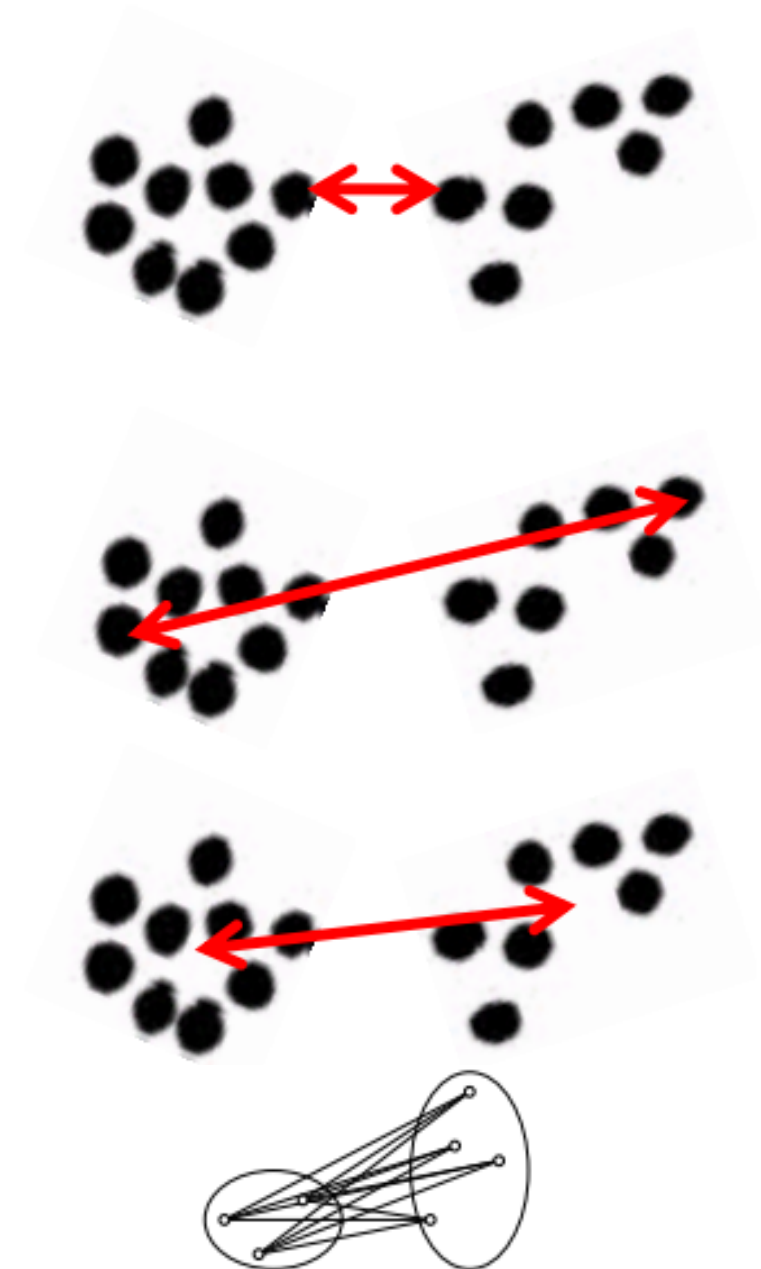
How to Define Distance Between Two Clusters?



Bottom-up Agglomerative clustering

Different algorithms differ in how the similarities are defined (and hence updated) between two clusters

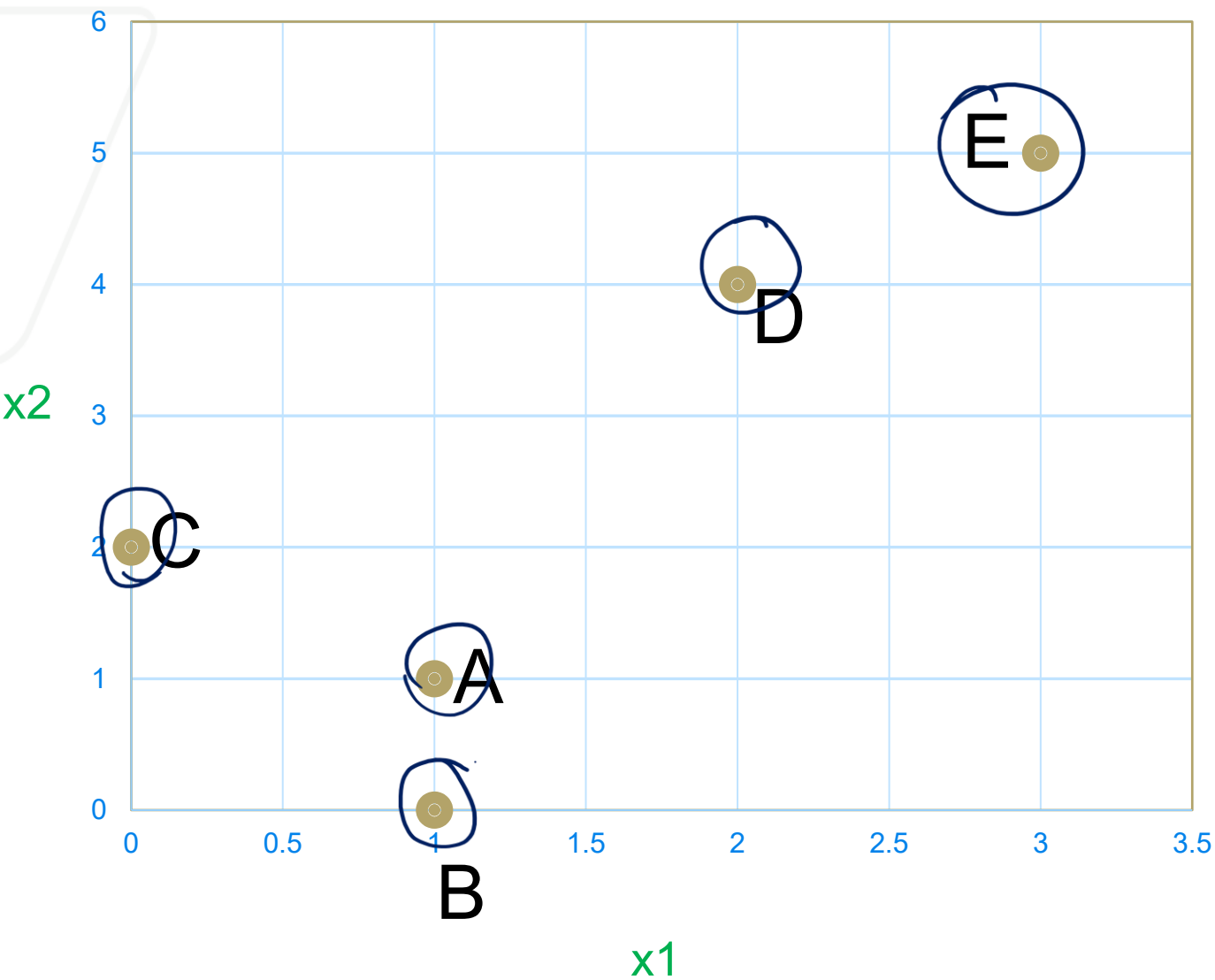
- Single-Link
 - Nearest Neighbor: similarity between their closest members.
- Complete-Link
 - Furthest Neighbor: similarity between their furthest members.
- Centroid
 - Similarity between the centers of gravity
- Average-Link
 - Average similarity of all cross-cluster pairs.



Distance Between Clusters

Different distance functions can lead to different results!

i	X1	X2
A	1	1
B	1	0
C	0	2
D	2	4
E	3	5



Calculate distance
between each
pair clusters.

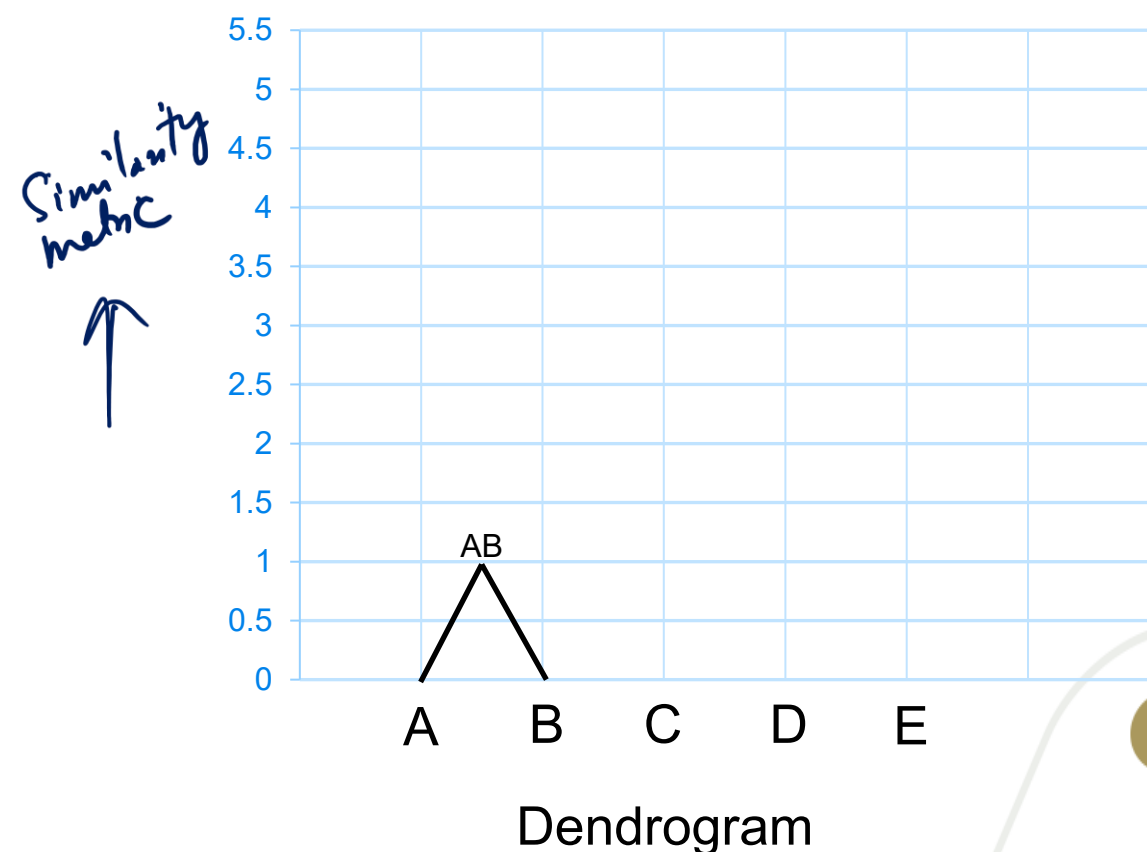
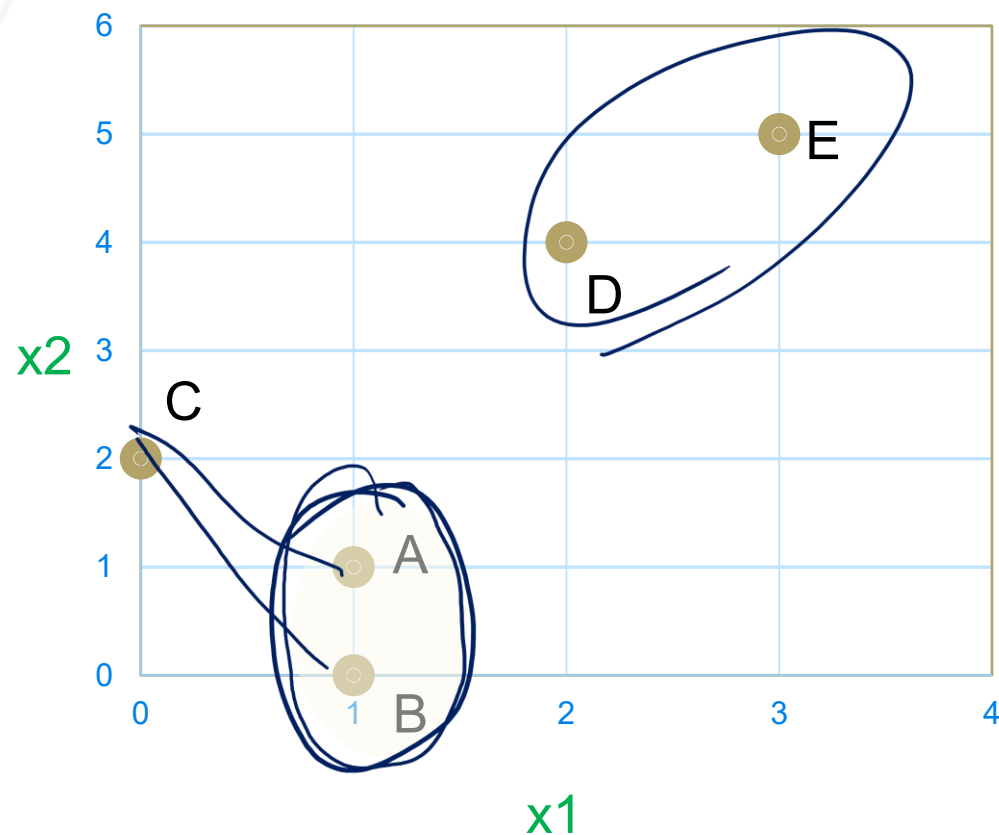
EUCLIDEAN DISTANCE

	A	B	C	D	E
A	0	1	1.4	3.2	4.5
B	1	0	2.2	4.1	5.4
C	1.4	2.2	0	2.8	4.2
D	3.2	4.1	2.8	0	1.4
E	4.5	5.4	4.2	1.4	0

Distance based on Average point (Bottom-Up Clustering)

	A	B	C	D	E
A	0	1	1.4	3.2	4.5
B	1	0	2.2	4.1	5.4
C	1.4	2.2	0	2.8	4.2
D	3.2	4.1	2.8	0	1.4
E	4.5	5.4	4.2	1.4	0

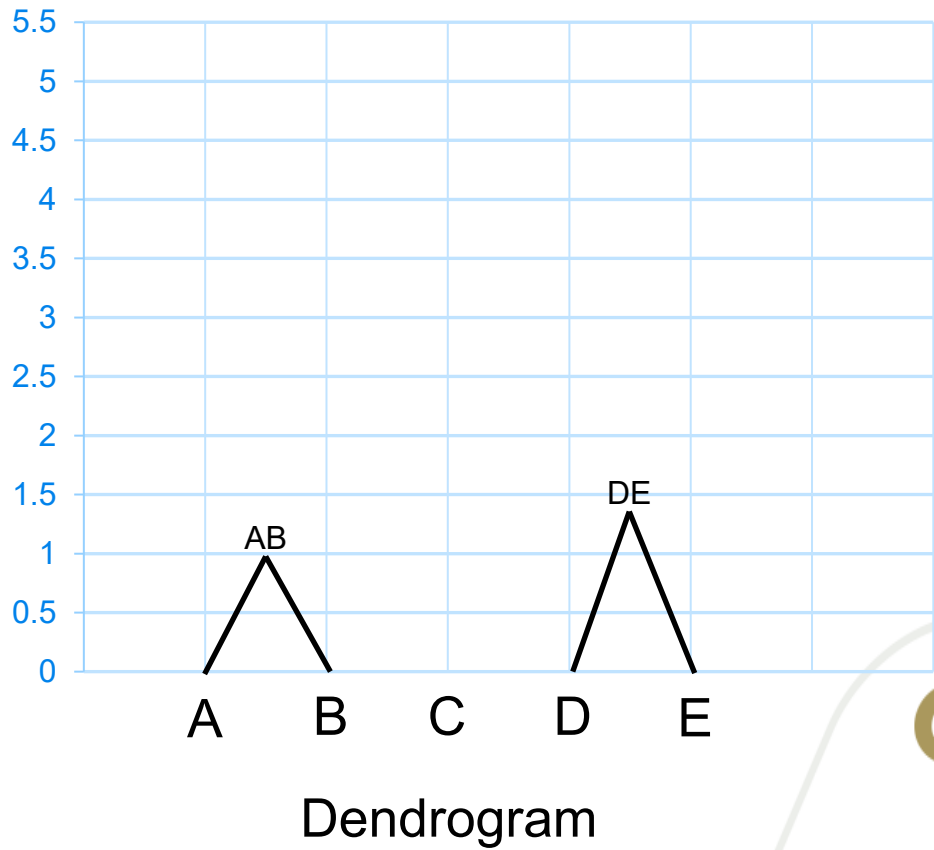
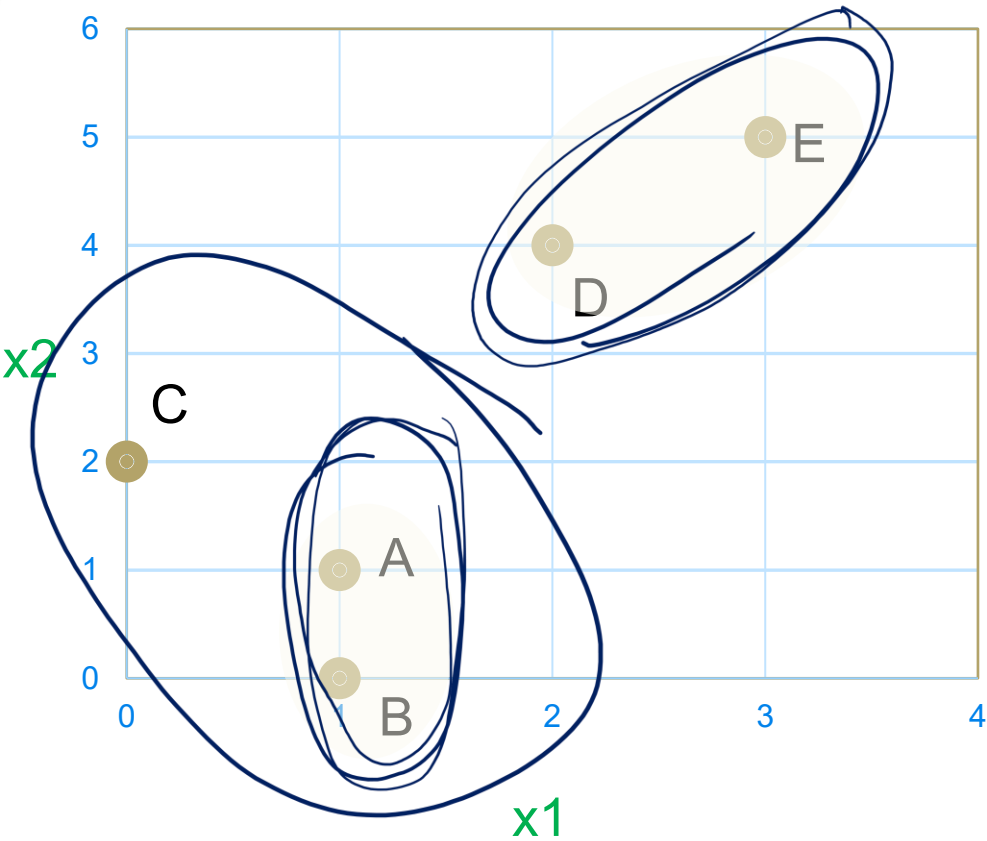
	(A,B)	C	D	E
(A,B)	0	1.8	3.6	4.9
C	1.8	0	2.8	4.2
D	3.6	2.8	0	1.4
E	4.9	4.2	1.4	0



Distance based on average point (Bottom-Up Clustering)

	(A,B)	C	D	E
(A,B)	0	1.8	3.6	4.9
C	1.8	0	2.8	4.2
D	3.6	2.8	0	1.4
E	4.9	4.2	1.4	0

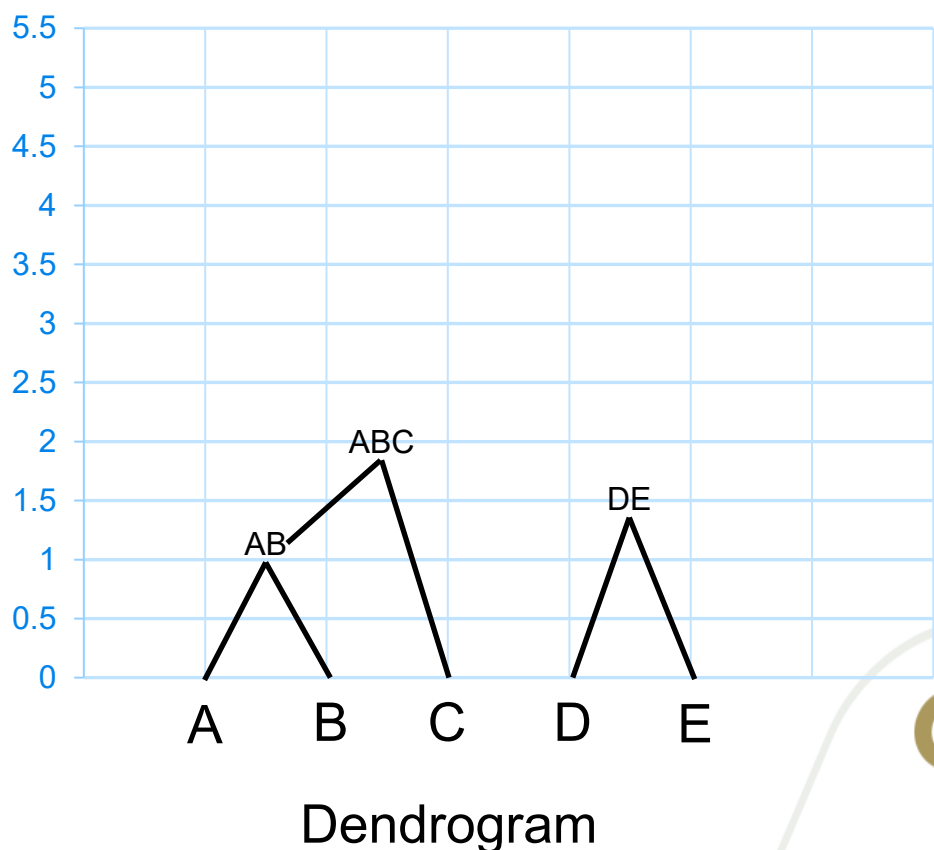
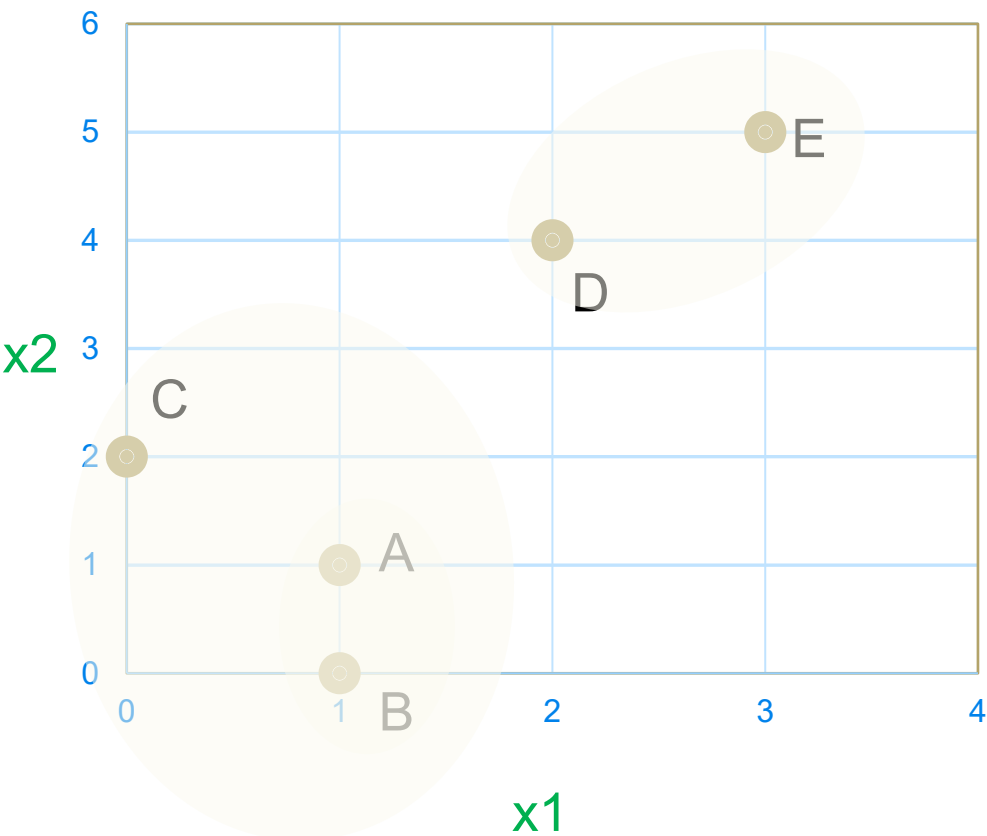
	(A,B)	C	(D,E)
(A,B)	0	1.8	4.25
C	1.8	0	3.5
(D,E)	4.25	3.5	0



Distance based on average point (Bottom-Up Clustering)

	(A,B)	C	(D,E)
(A,B)	0	1.8	4.25
C	1.8	0	3.5
(D,E)	4.25	3.5	0

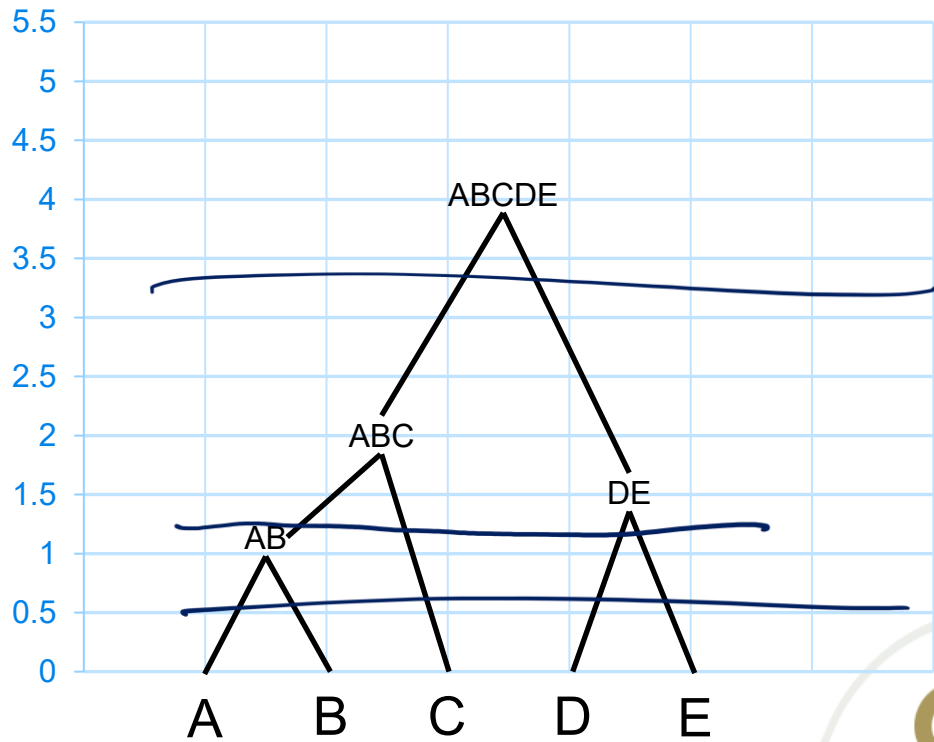
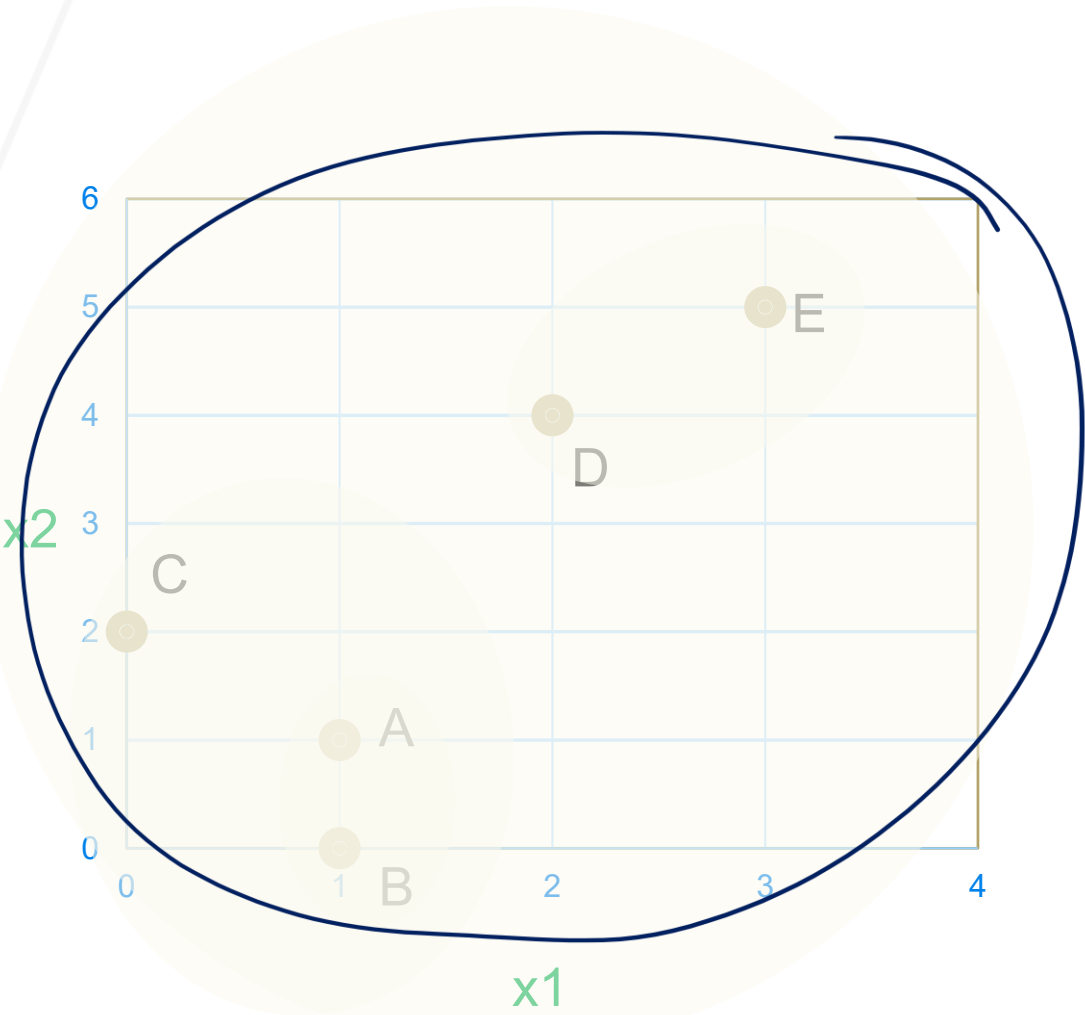
	((A,B),C)	(D,E)
((A,B),C)	0	3.875
(D,E)	3.875	0



Distance based on average point (Bottom-Up Clustering)

	((A,B),C)	(D,E)
((A,B),C)	0	3.875
(D,E)	3.875	0

	(((A,B),C),(D,E))
(((A,B),C),(D,E))	0

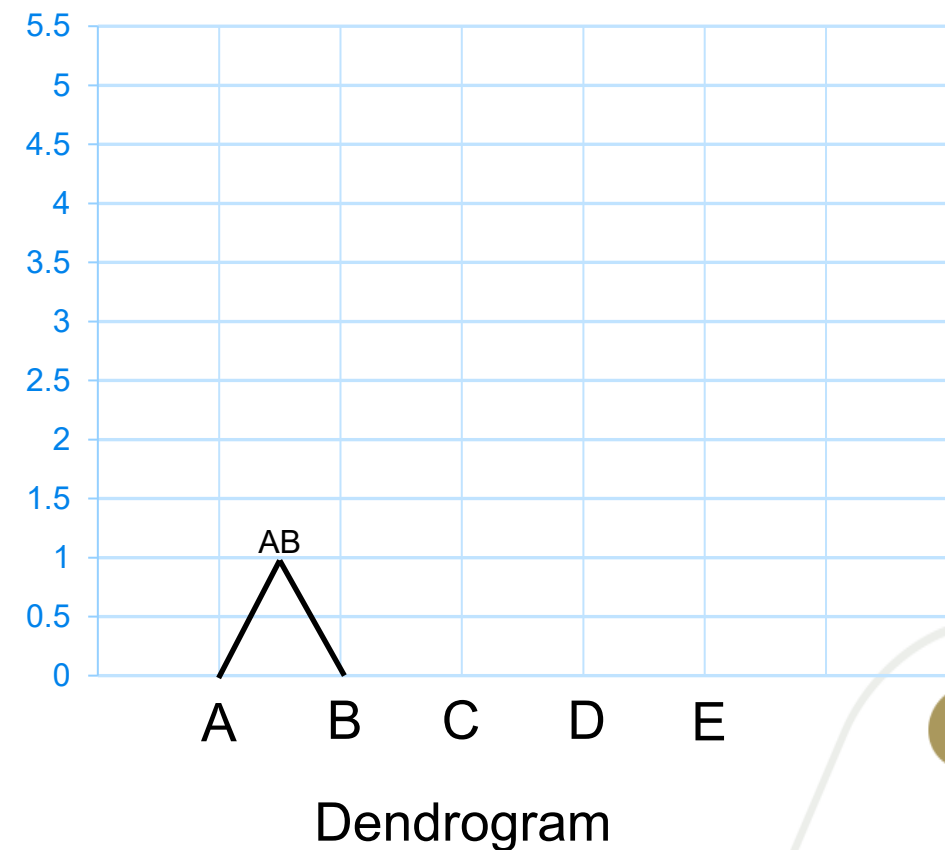
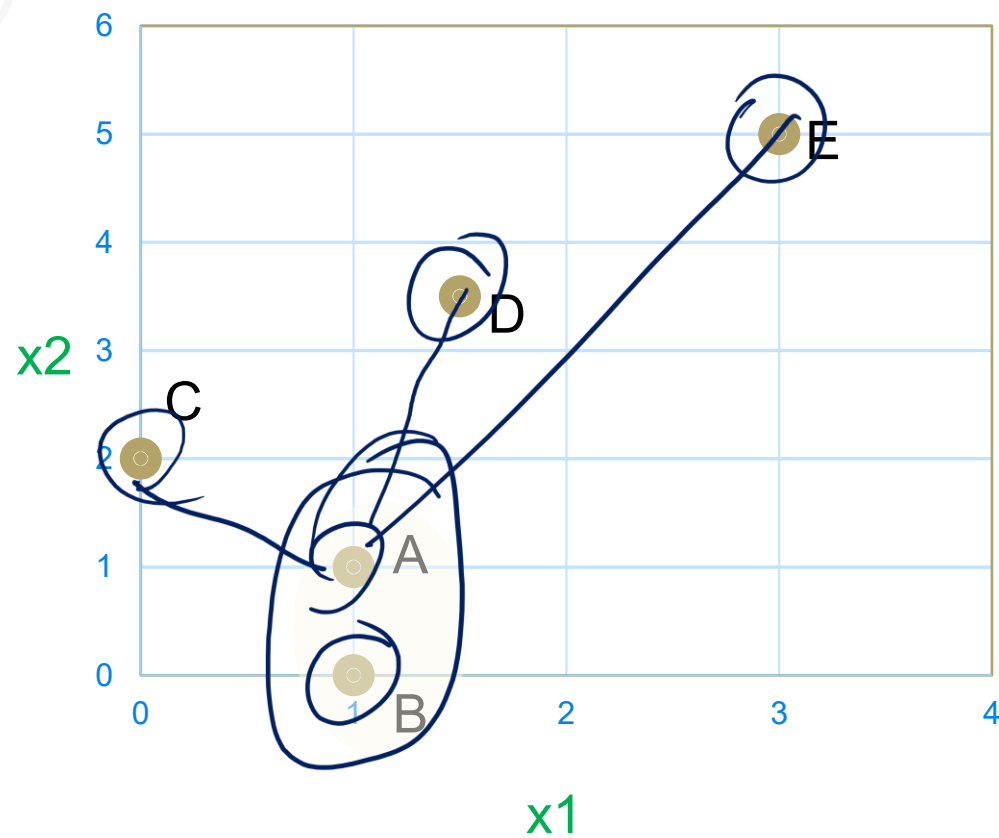


Dendrogram

Distance based on Single Link (Bottom-Up Clustering)

	A	B	C	D	E
A	0	1	1.4	2.55	4.5
B	1	0	2.2	3.53	5.4
C	1.4	2.2	0	2.12	4.2
D	2.55	3.53	2.12	0	2.12
E	4.5	5.4	4.2	2.12	0

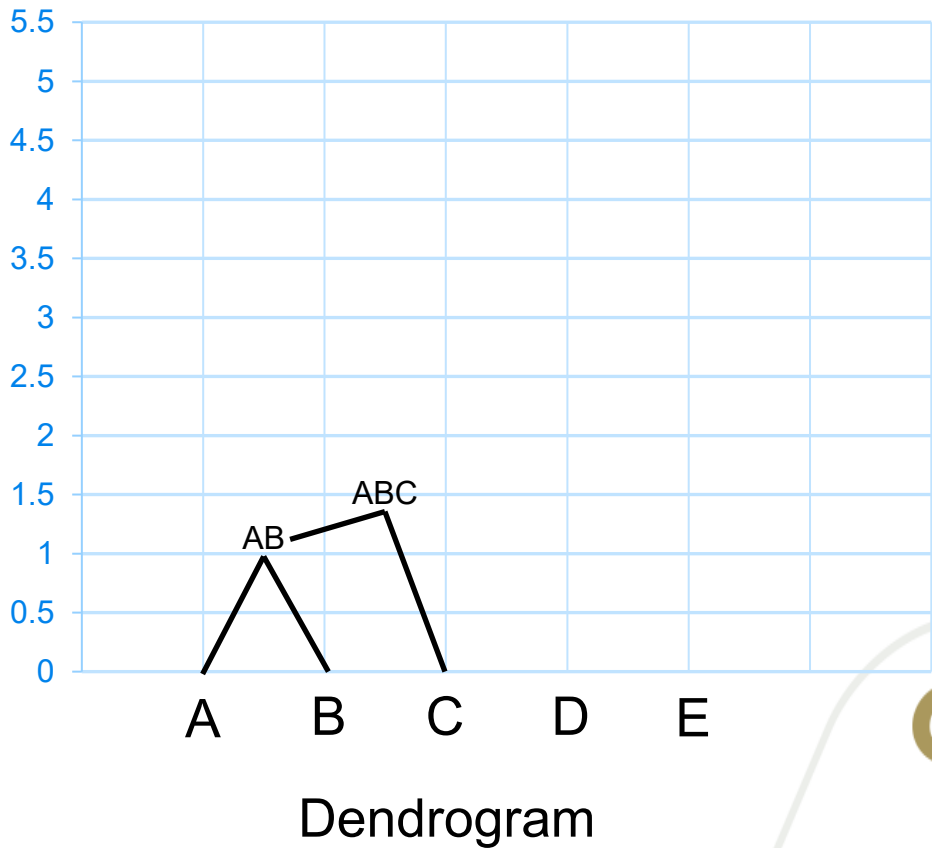
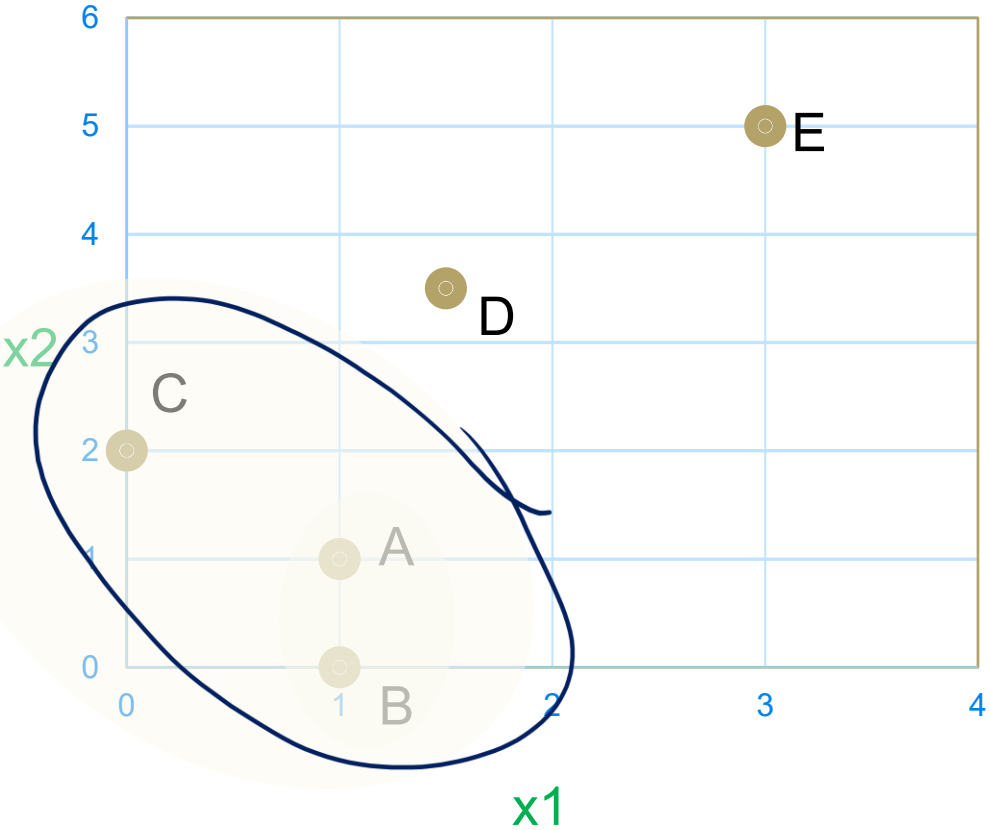
	(A,B)	C	D	E
(A,B)	0	1.4	2.55	4.5
C	1.4	0	2.12	4.2
D	2.55	2.12	0	2.12
E	4.5	4.2	2.12	0



Distance based on Single Link (Bottom-Up Clustering)

	(A,B)	C	D	E
(A,B)	0	1.4	2.55	4.5
C	1.4	0	2.12	4.2
D	2.55	2.12	0	2.12
E	4.5	4.2	2.12	0

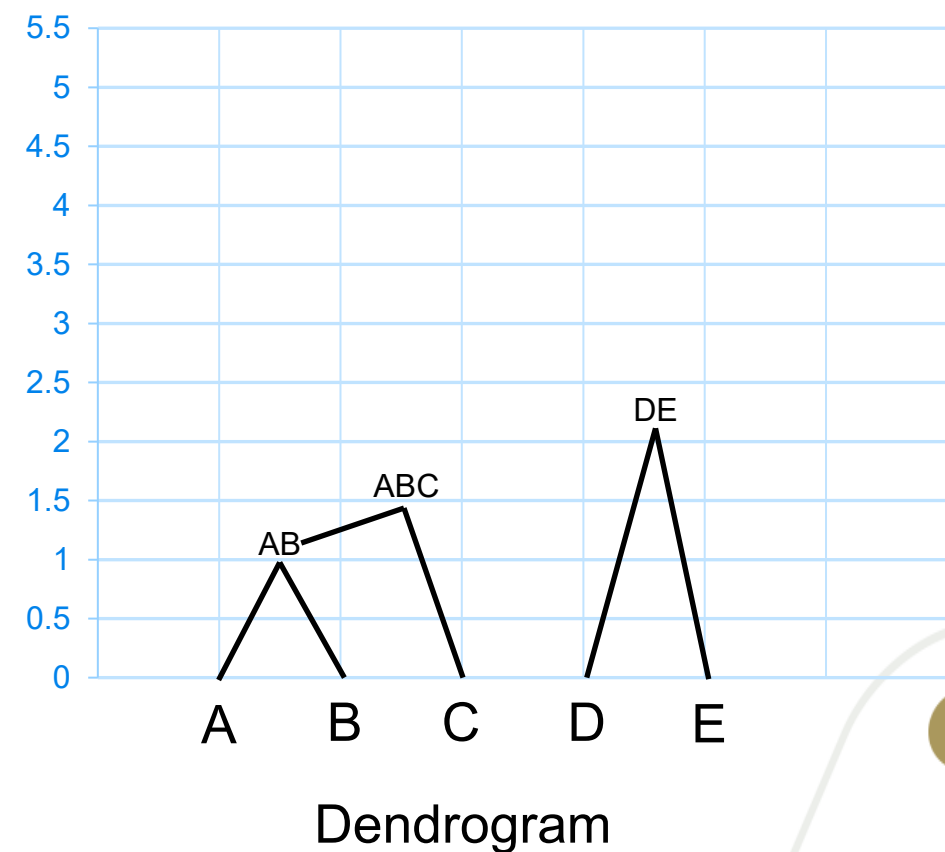
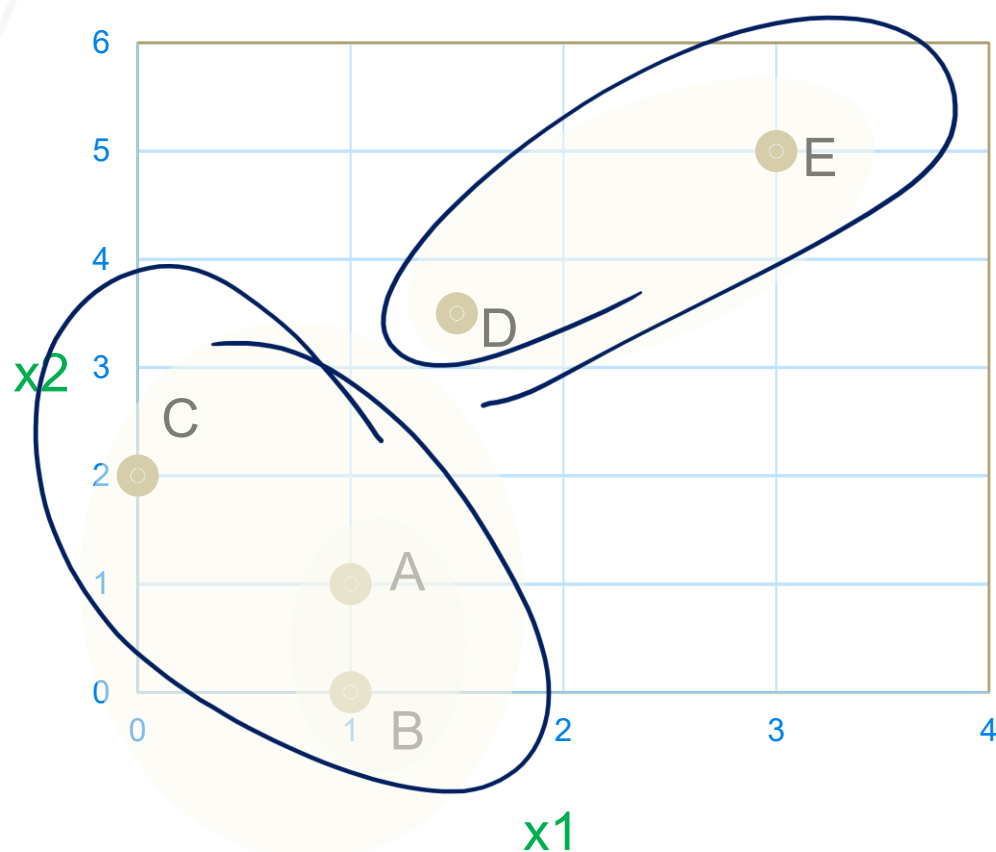
	(A,B),C	D	E
(A,B),C	0	<u>2.12</u>	4.2
D	<u>2.12</u>	0	<u>2.12</u>
E	4.2	2.12	0



Distance based on Single Link (Bottom-Up Clustering)

	(A,B), C	D	E
(A,B), C	0	2.12	4.2
D	2.12	0	2.12
E	4.2	2.12	0

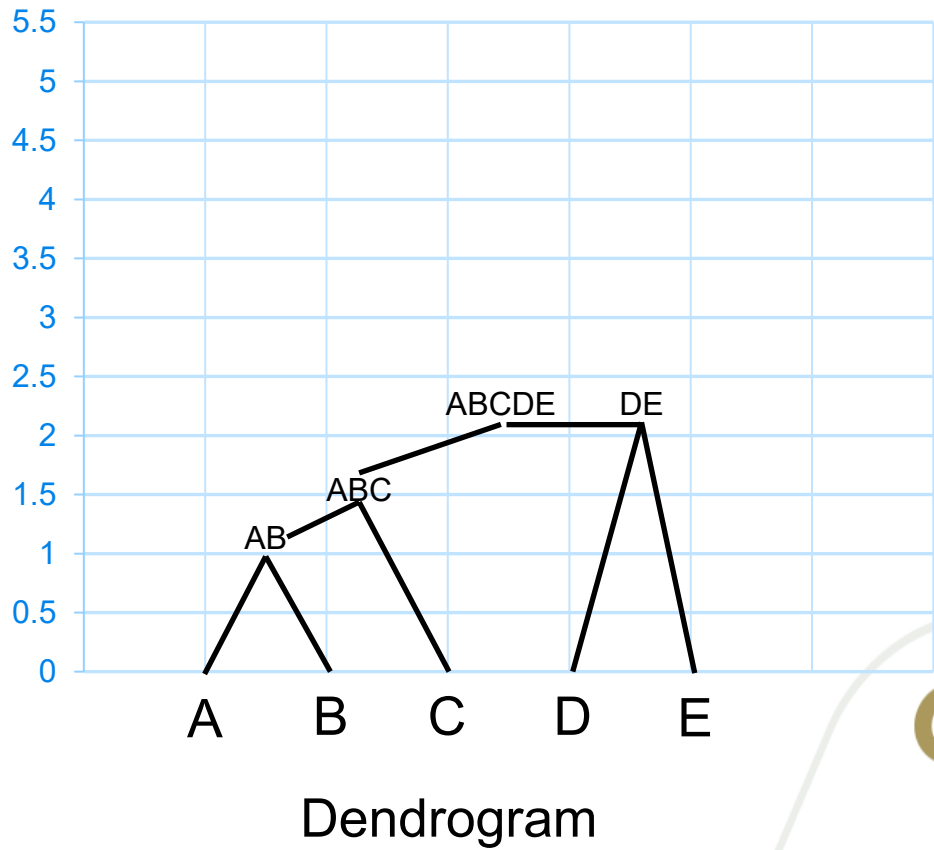
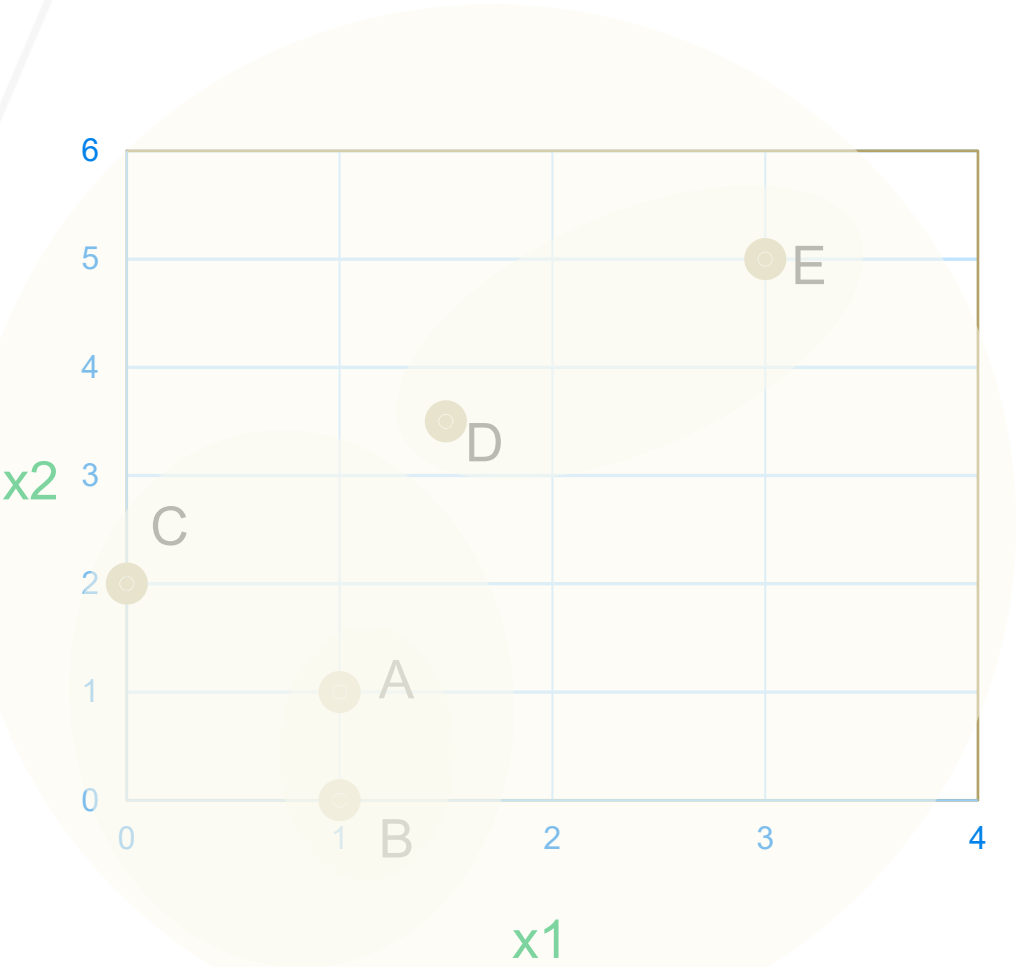
	((A,B),C)	(D,E)
((A,B),C)	0	2.12
(D,E)	2.12	0



Distance based on Single Link (Bottom-Up Clustering)

	$((A,B),C)$	(D,E)
$((A,B),C)$	0	2.12
(D,E)	2.12	0

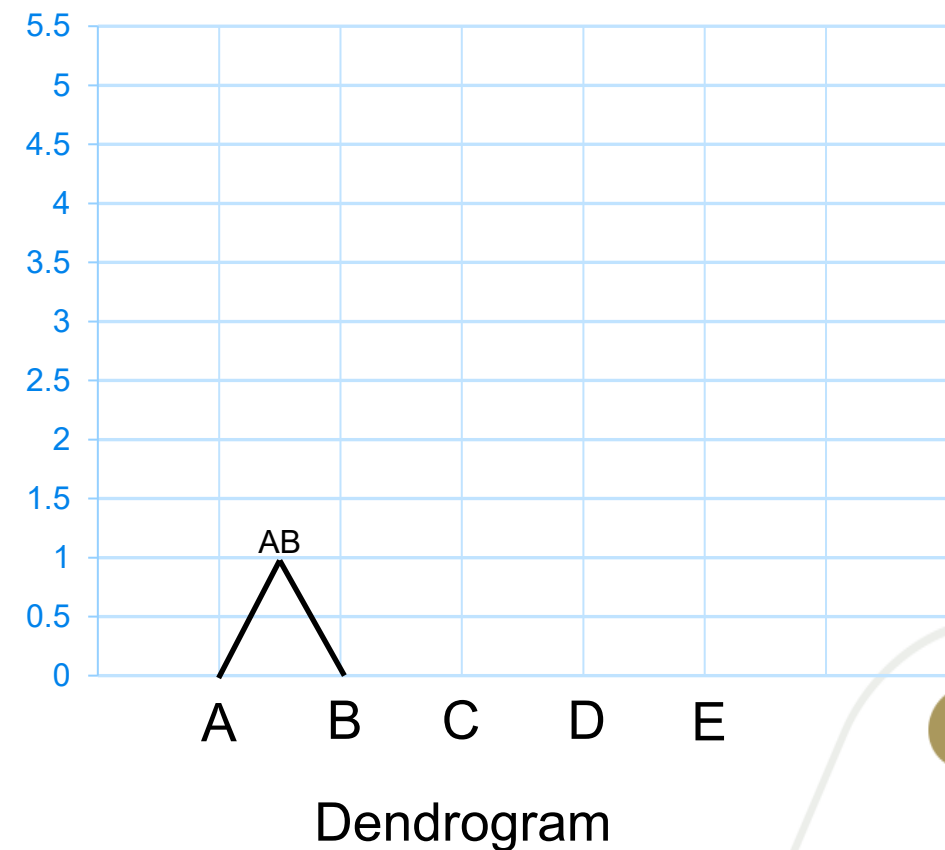
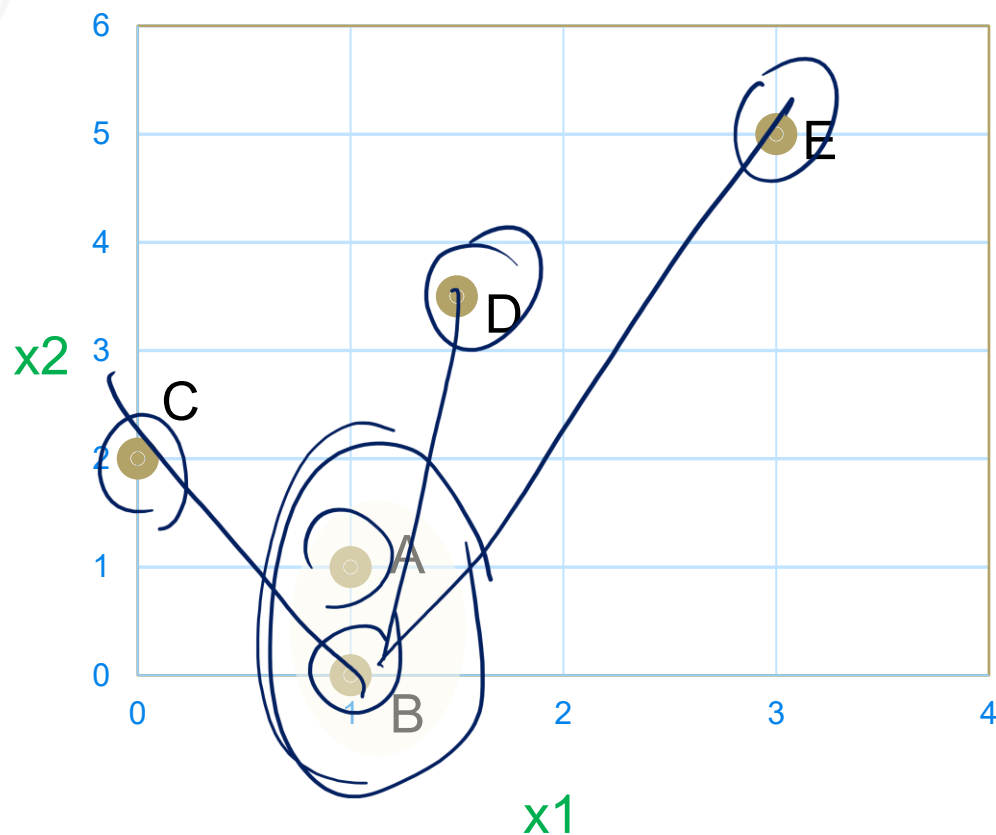
	$((((A,B),C),D),E)$
$((((A,B),C),D),E)$	0



Distance based on Complete Link (Bottom-Up Clustering)

	A	B	C	D	E
A	0	1	1.4	2.55	4.5
B	1	0	2.2	3.53	5.4
C	1.4	2.2	0	2.12	4.2
D	2.55	3.53	2.12	0	2.12
E	4.5	5.4	4.2	2.12	0

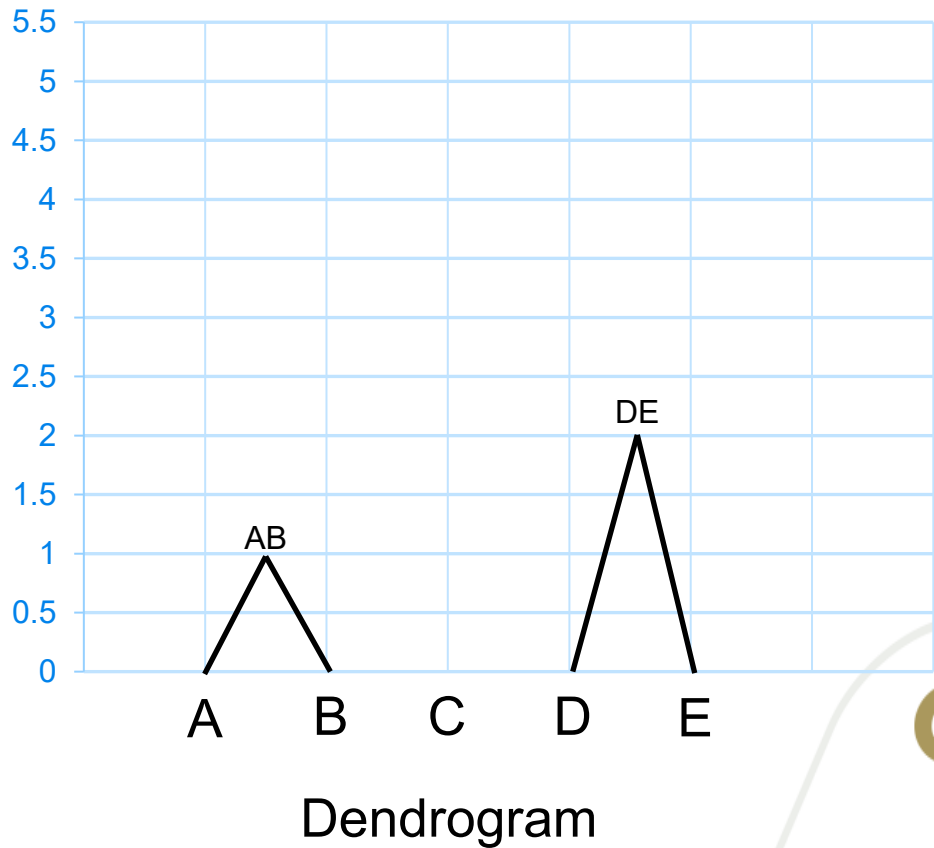
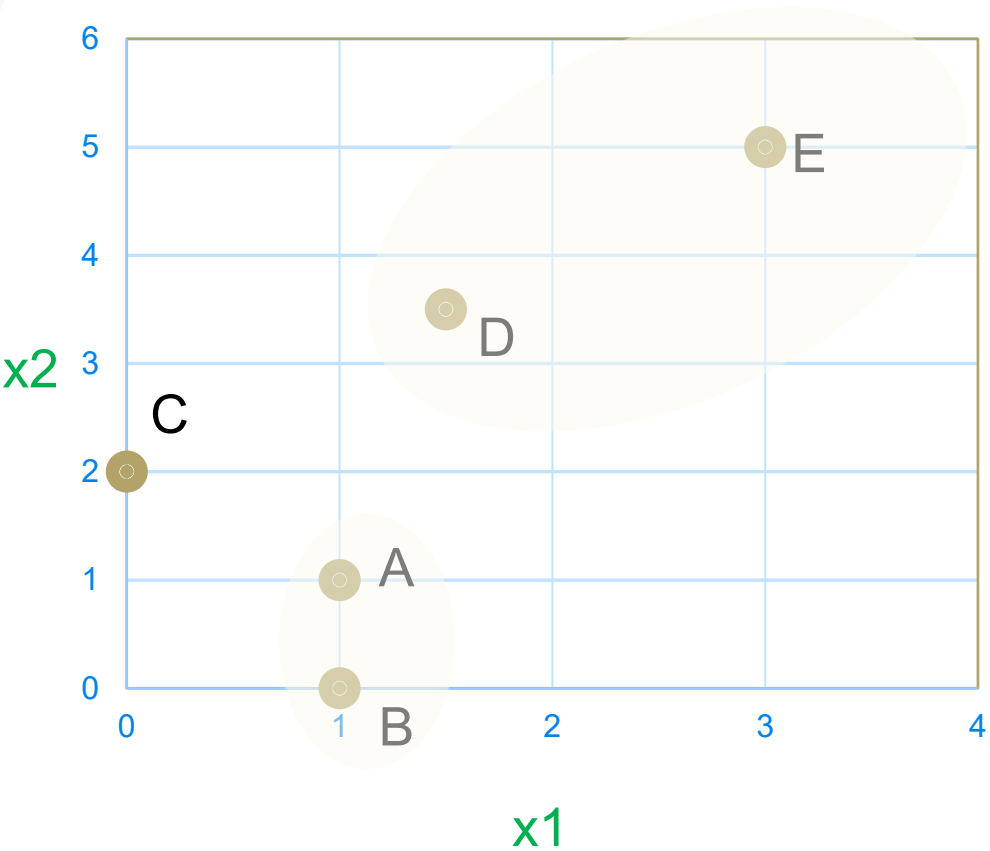
	(A,B)	C	D	E
(A,B)	0	2.2	3.53	5.4
C	2.2	0	2.12	4.2
D	3.55	2.12	0	2.12
E	5.4	4.2	2.12	0



Distance based on Complete Link (Bottom-Up Clustering)

	(A,B)	C	D	E
(A,B)	0	2.2	3.55	5.4
C	2.2	0	2.12	4.2
D	3.55	2.12	0	2.12
E	5.4	4.2	2.12	0

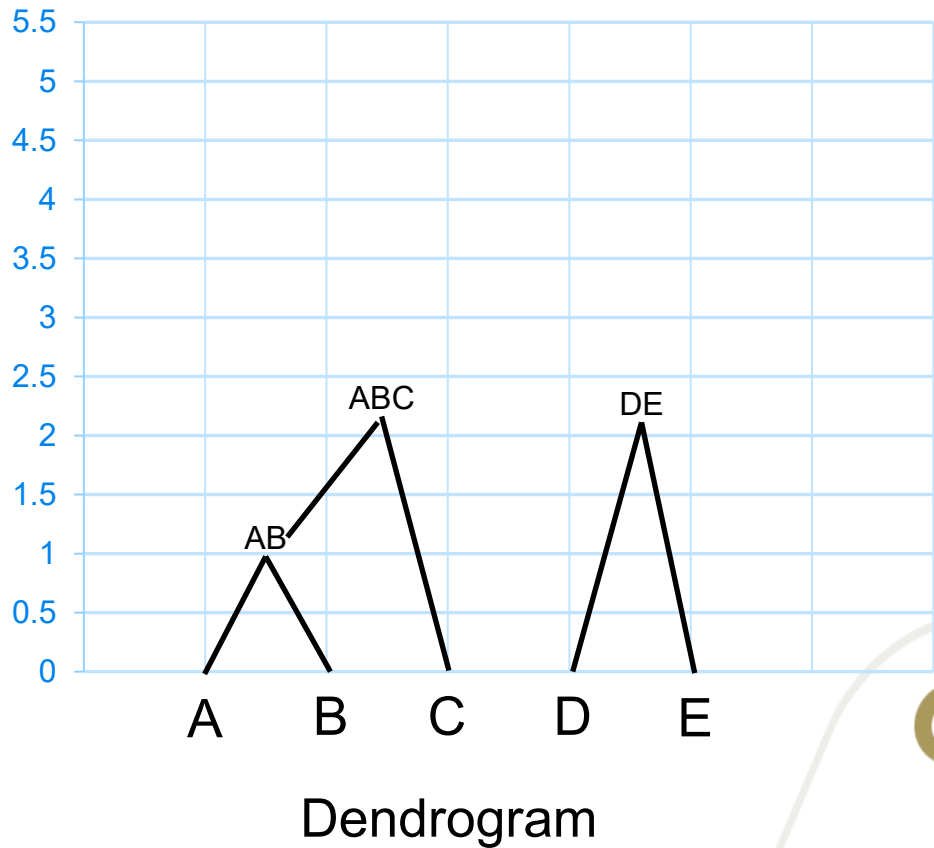
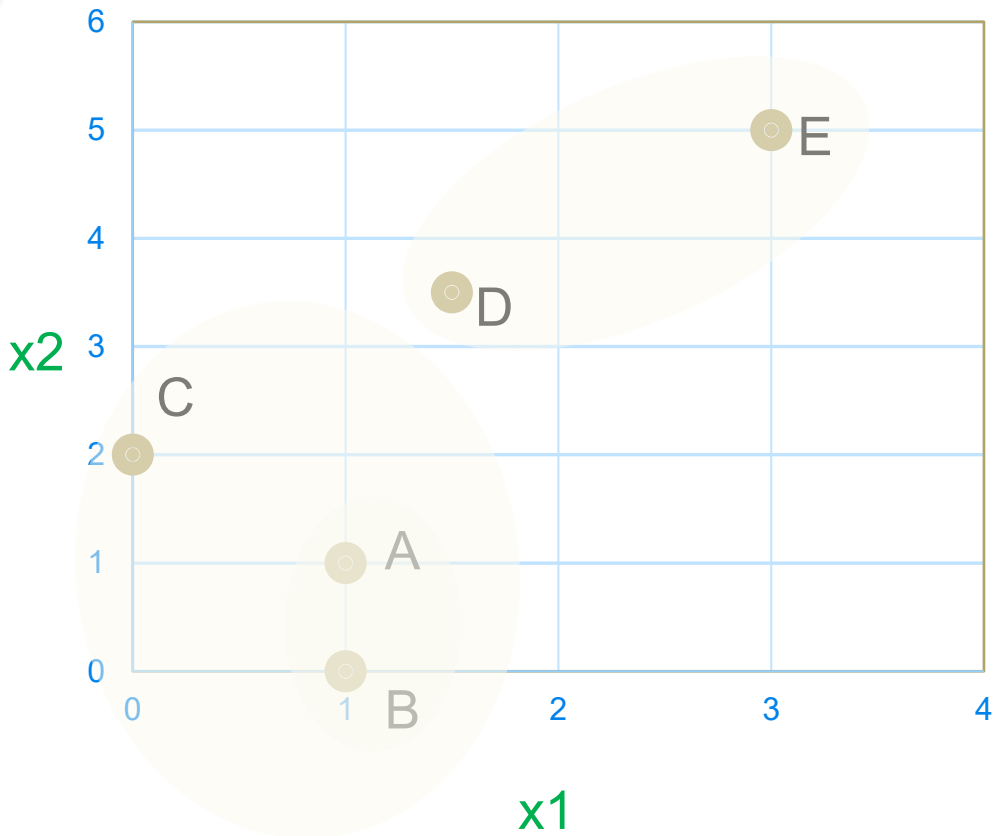
	(A,B)	C	(D,E)
(A,B)	0	2.2	5.4
C	2.2	0	4.2
(D,E)	5.4	4.2	0



Distance based on Single Link (Bottom-Up Clustering)

	(A,B)	C	(D,E)
(A,B)	0	2.2	5.4
C	2.2	0	4.2
(D,E)	5.4	4.2	0

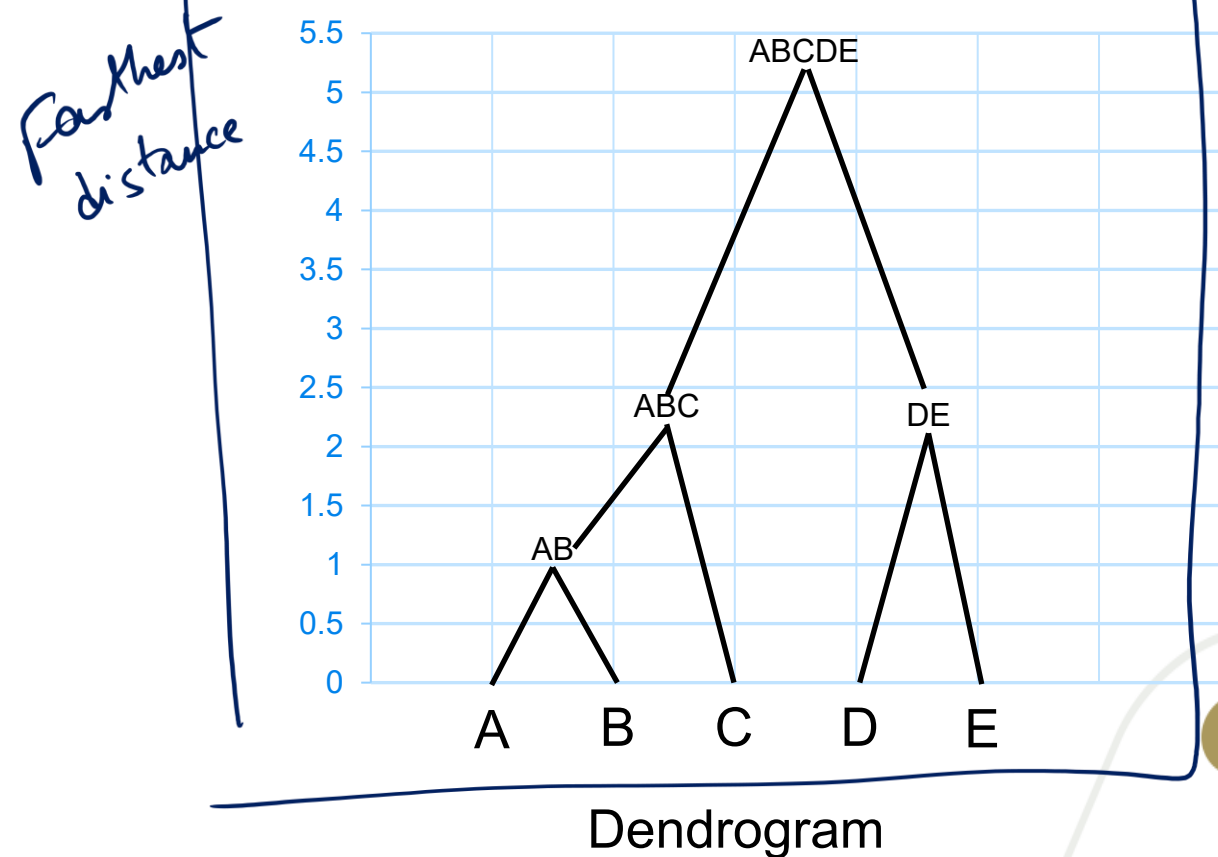
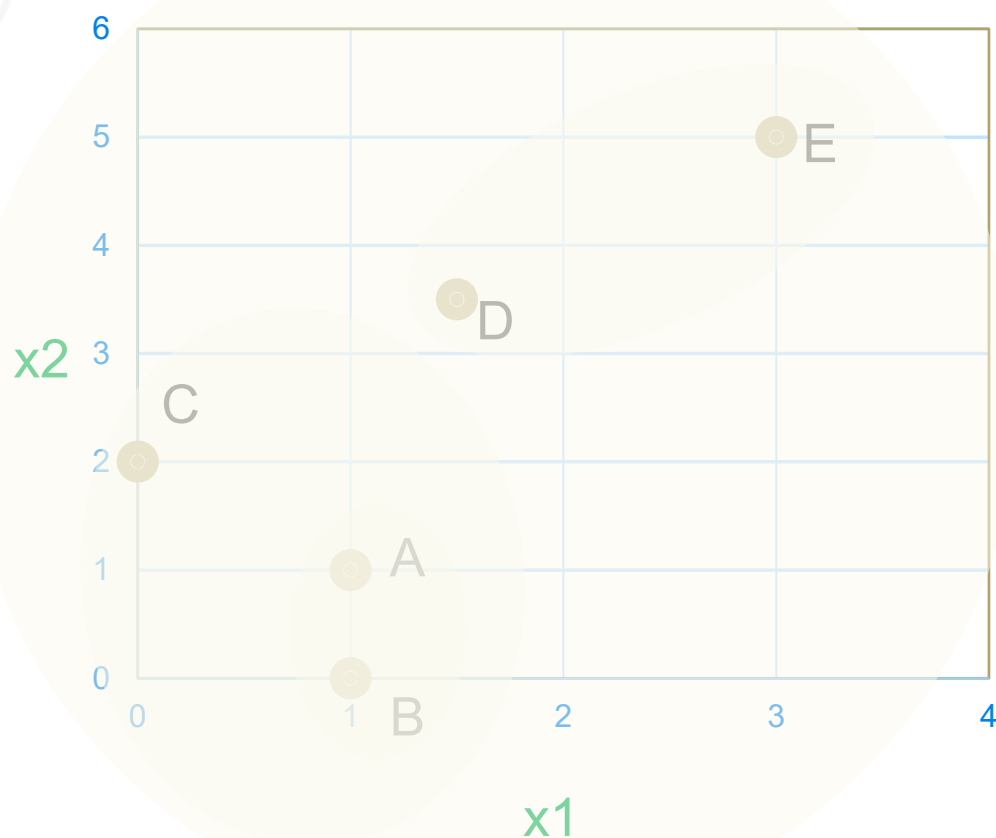
	((A,B),C)	(D,E)
((A,B),C)	0	5.4
(D,E)	5.4	0

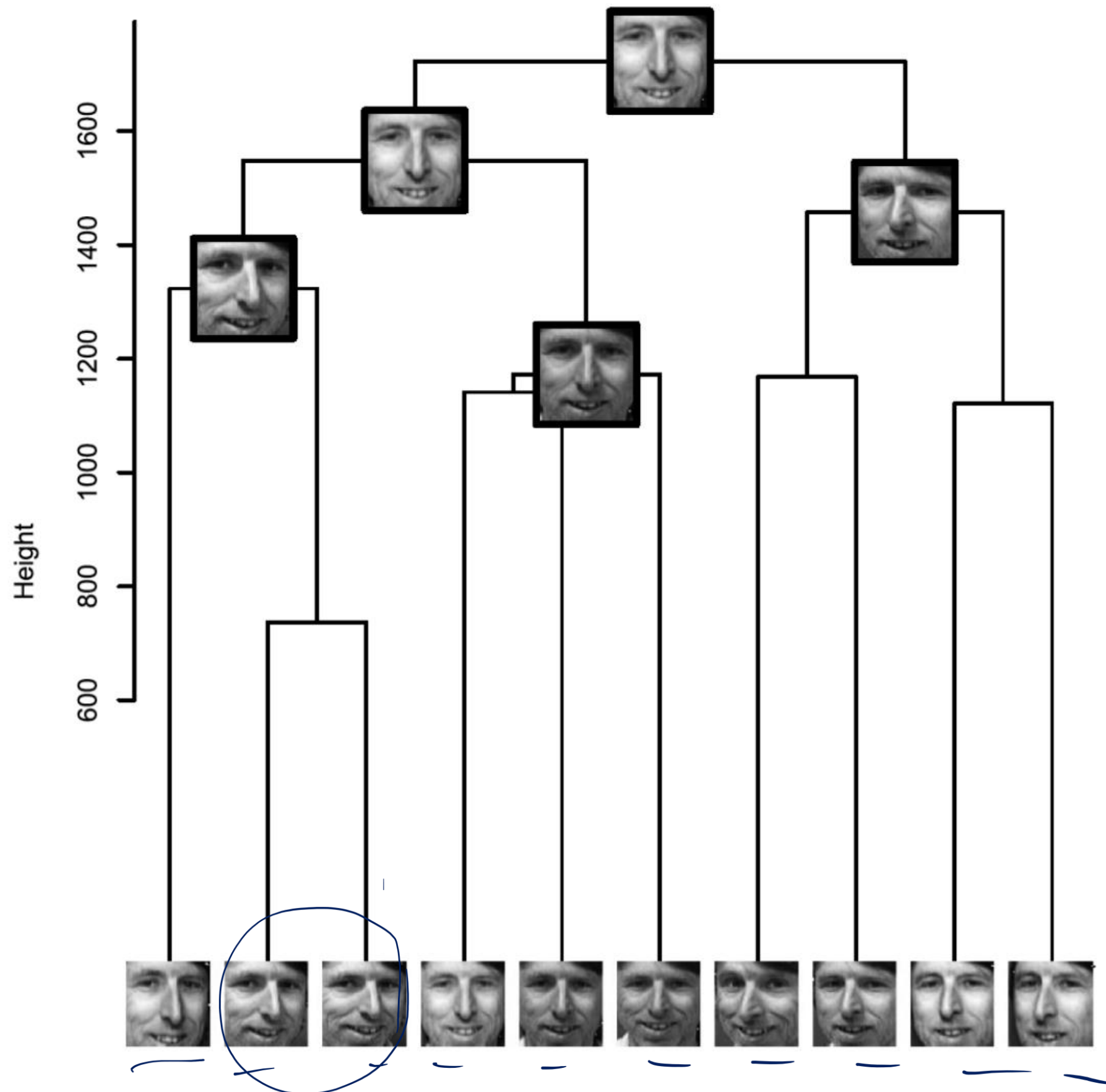


Distance based on Complete Link (Bottom-Up Clustering)

	$((A,B),C)$	(D,E)
$((A,B),C)$	0	5.4
(D,E)	5.4	0

	$((((A,B),C),D),E)$
$((((A,B),C),D),E)$	0

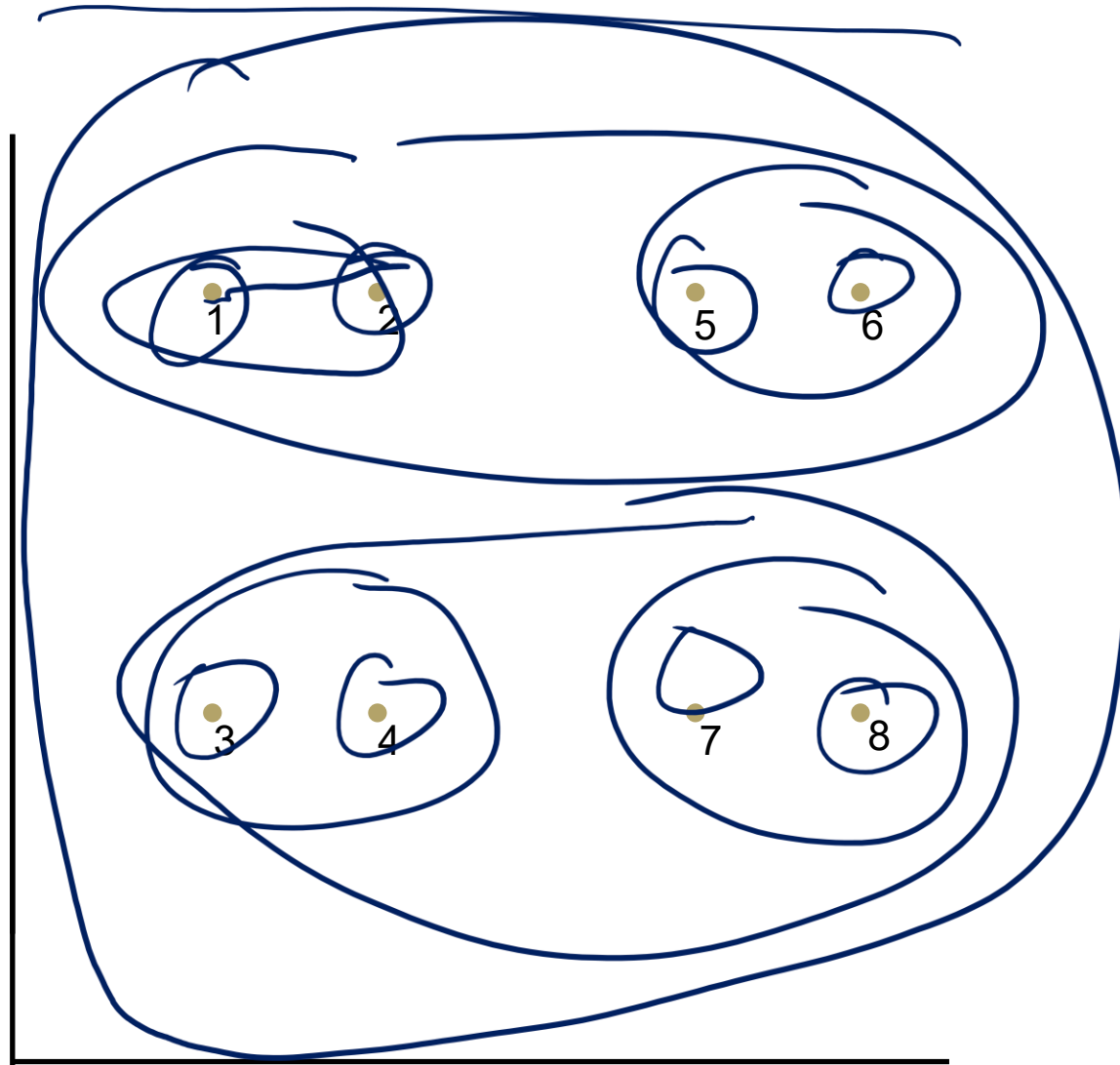




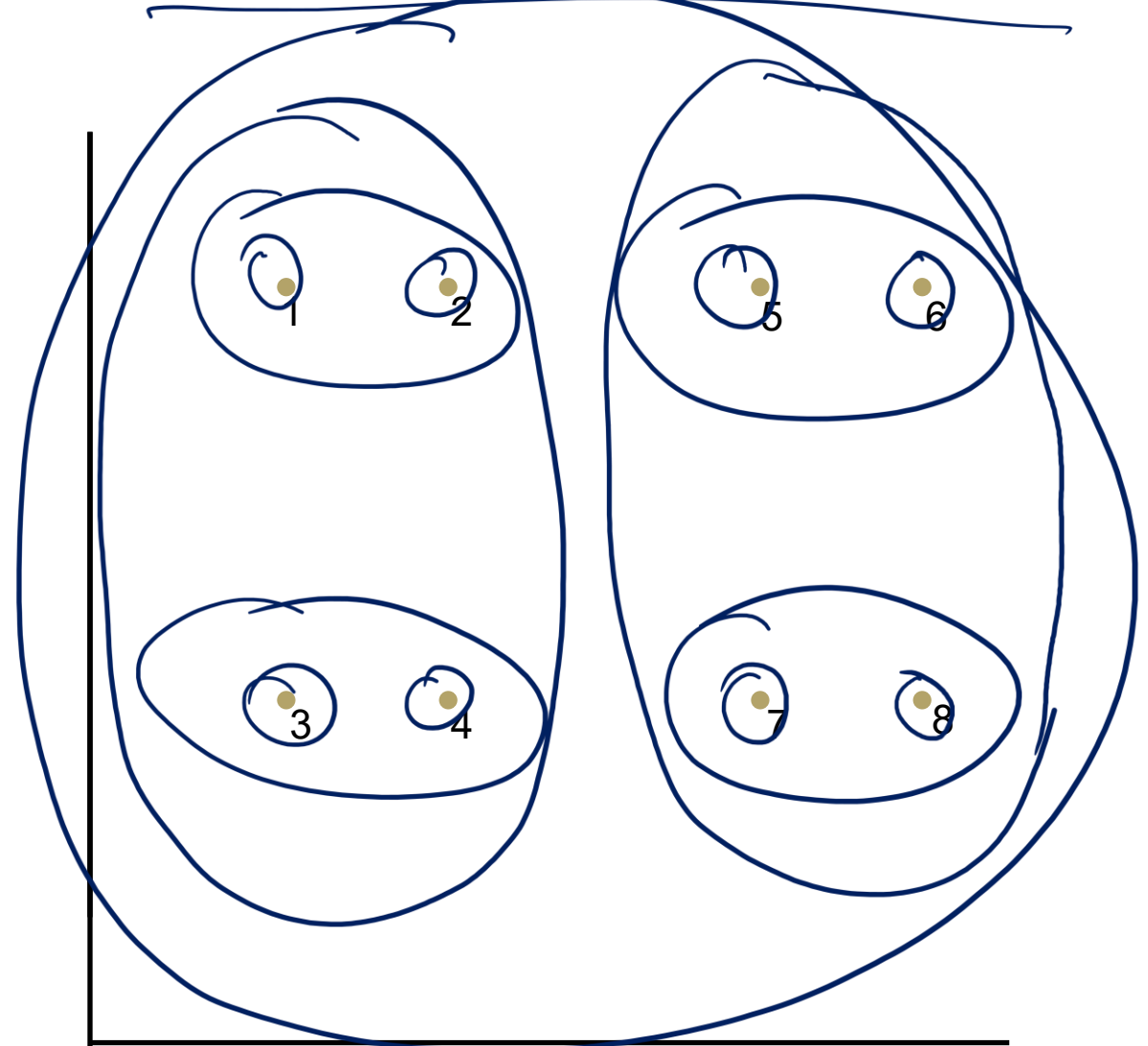
(From Bien et al. (2011))

Another Example

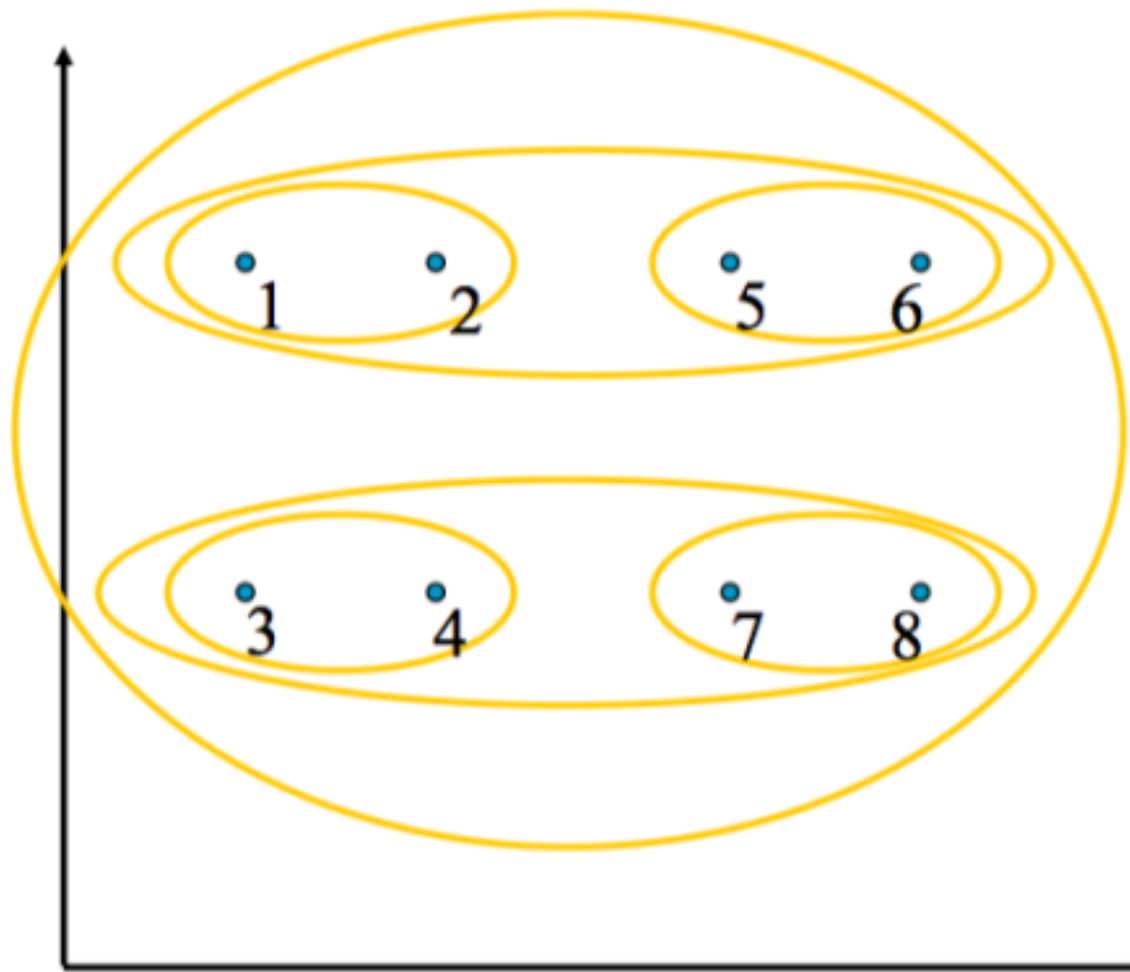
Single Link clustering (Closest pair)



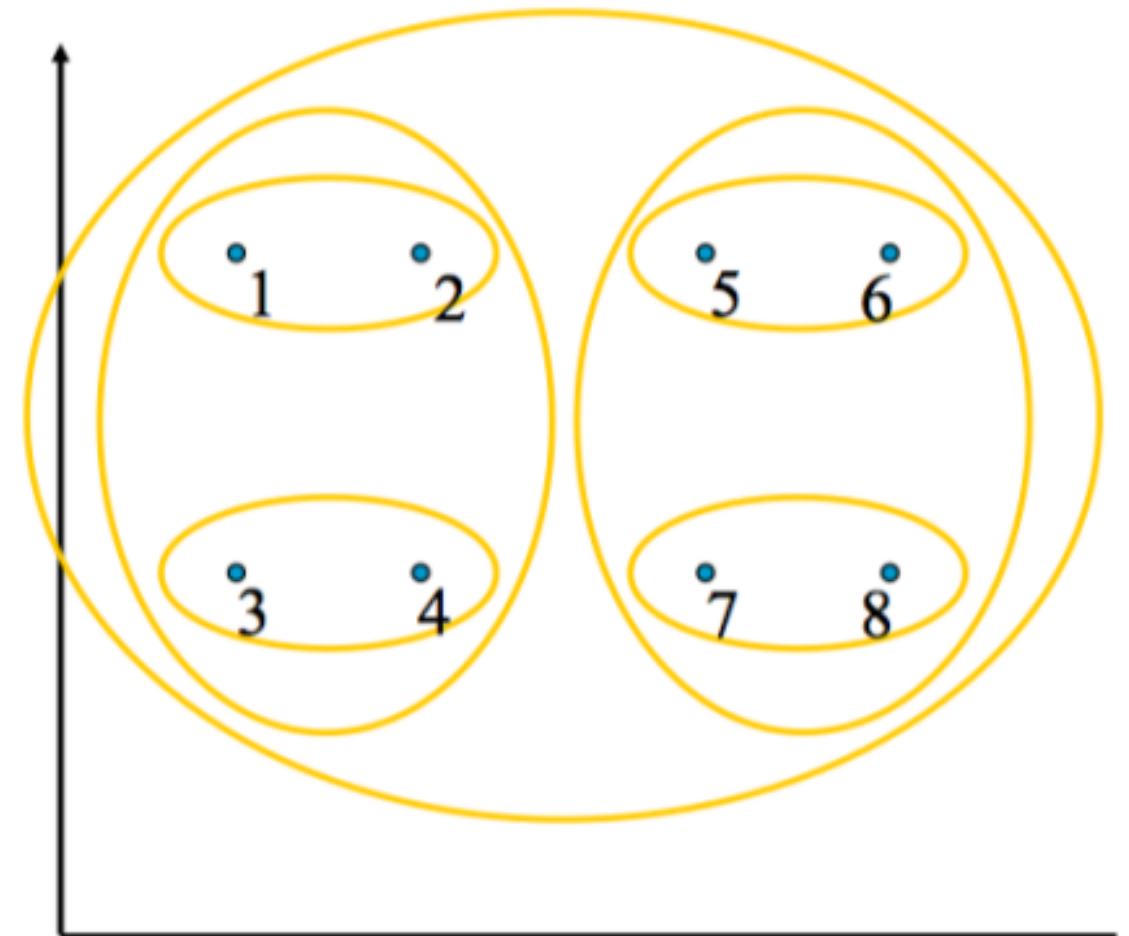
Complete Link clustering (Farthest pair)

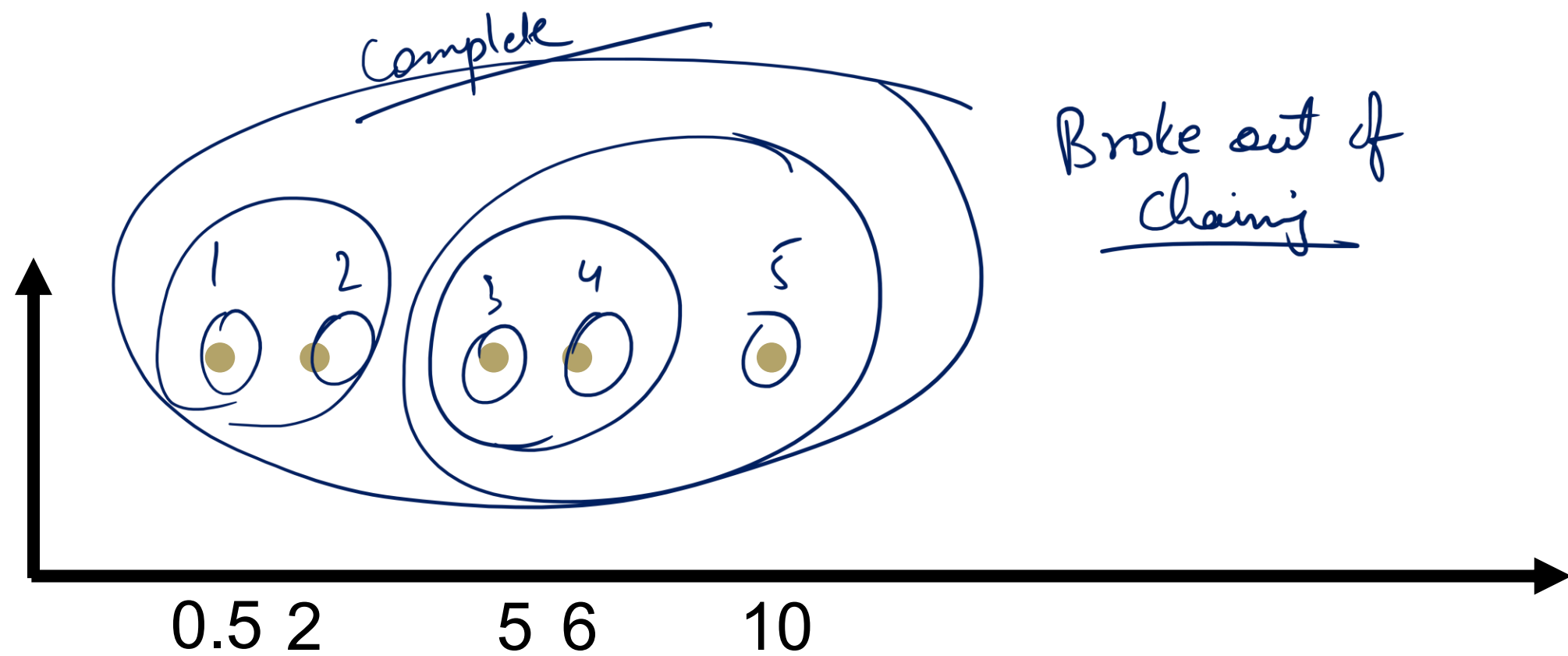
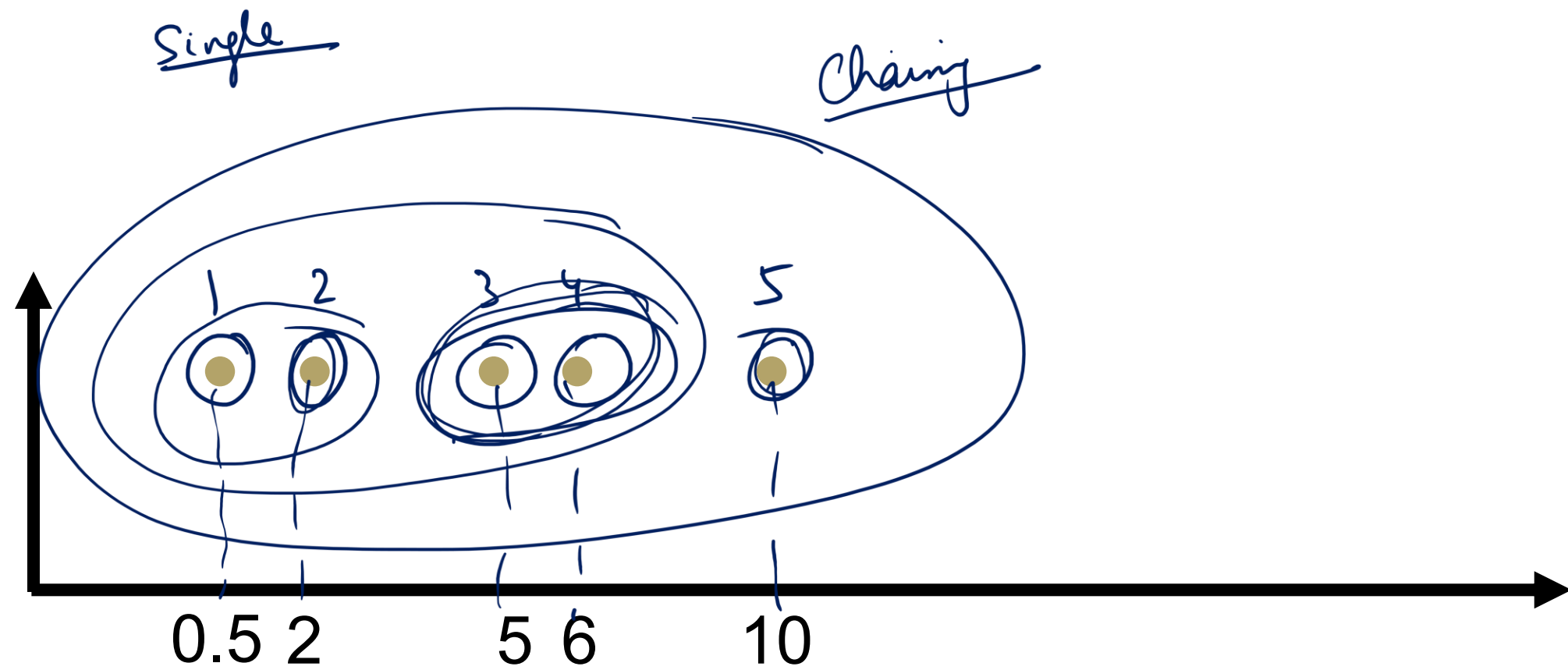


Closest pair (single-link clustering)

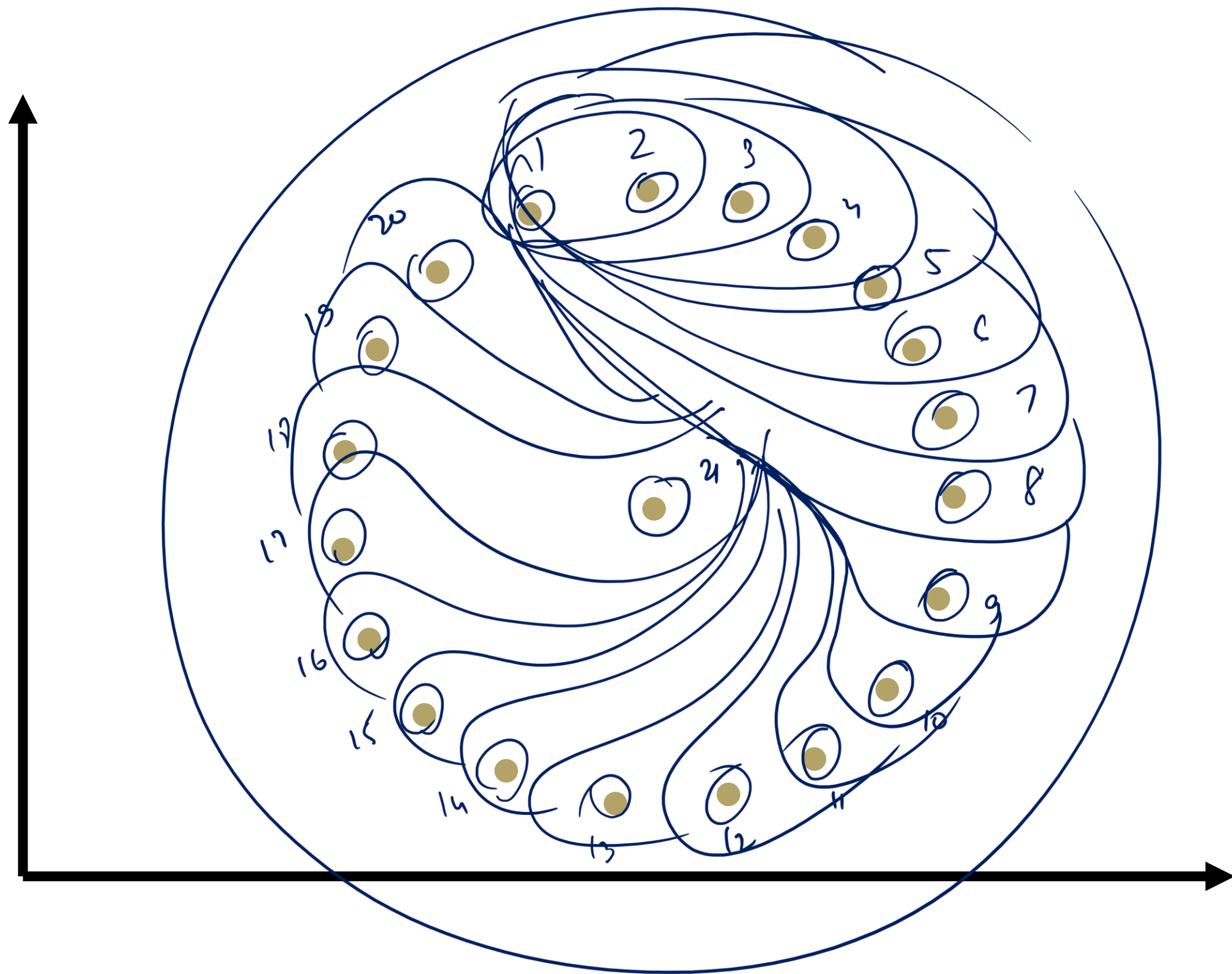


Farthest pair (complete-link clustering)

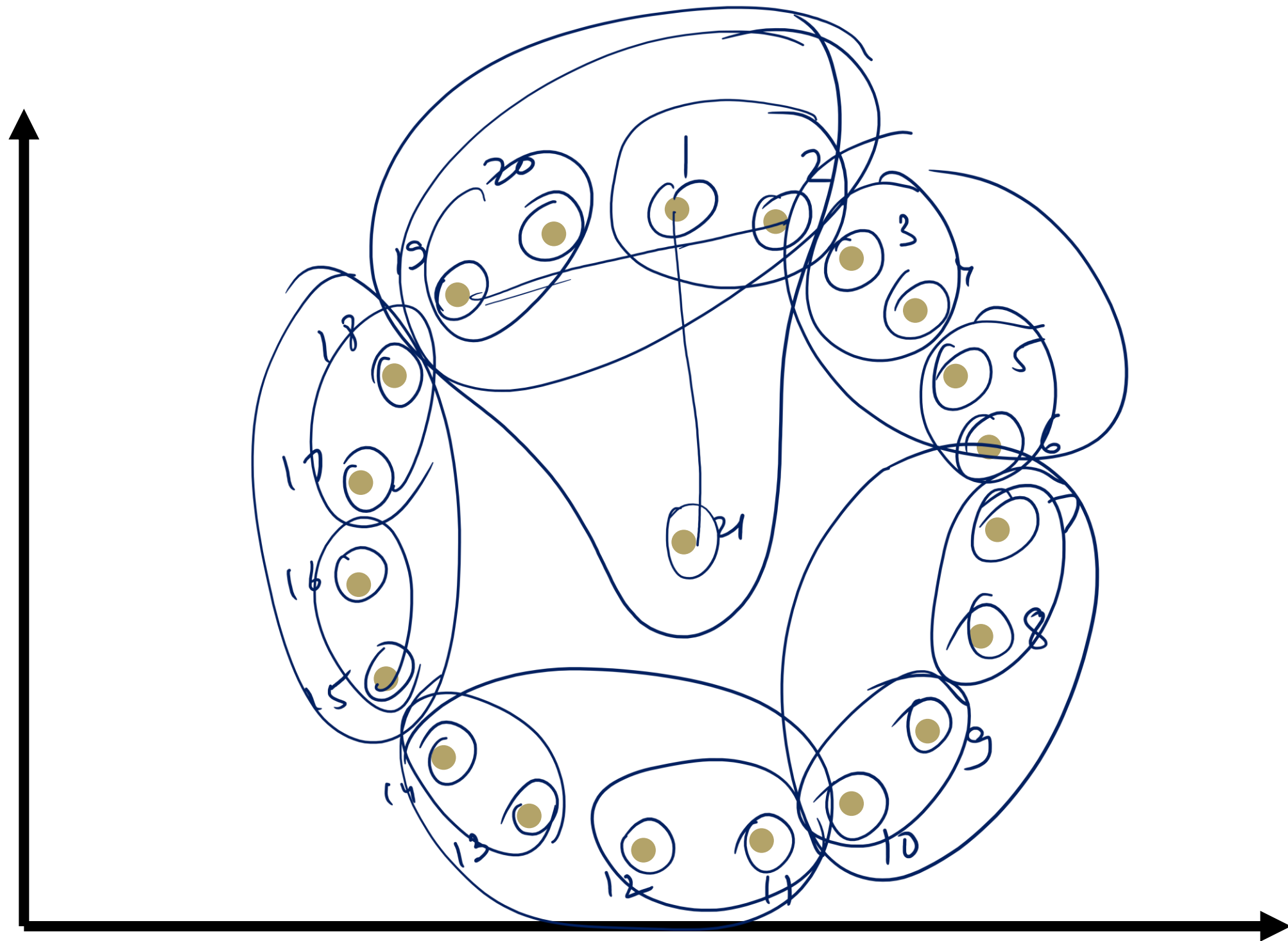




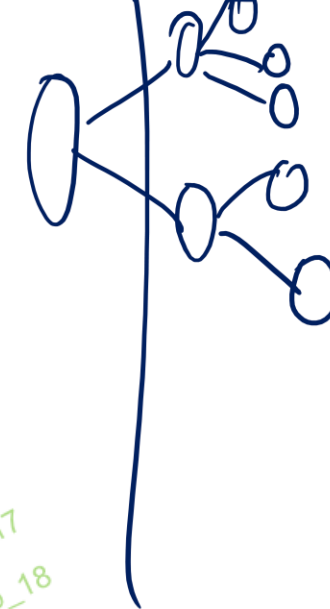
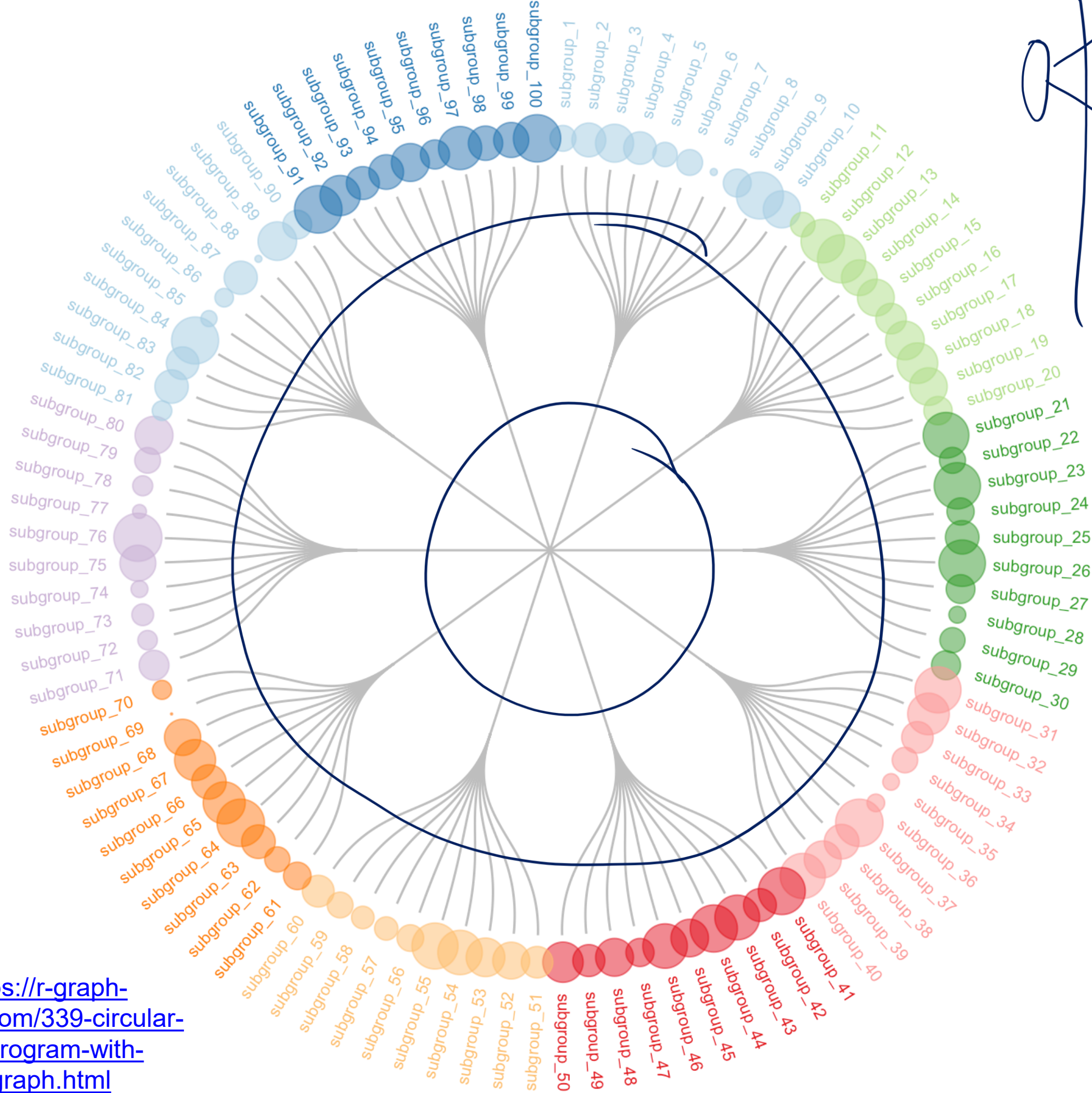
We like chaining → Single



Complete



<https://r-graph-gallery.com/339-circular-dendrogram-with-ggraph.html>



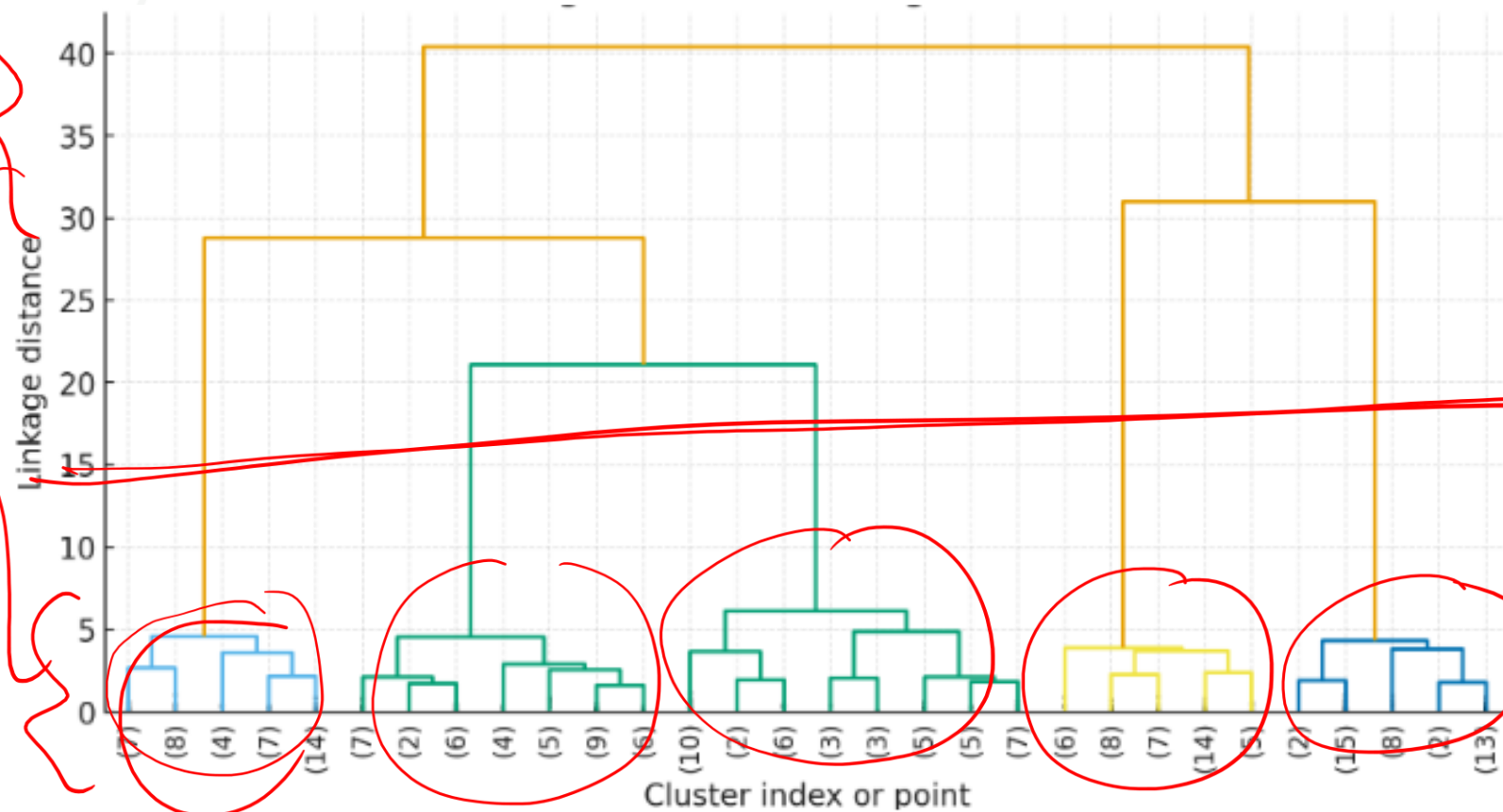
Clustering Evaluation

- Internal measures for clustering evaluation
 - Elbow method
 - Silhouette Coefficient
 - Graph-based measures (Beta-CV and Normalized cut)

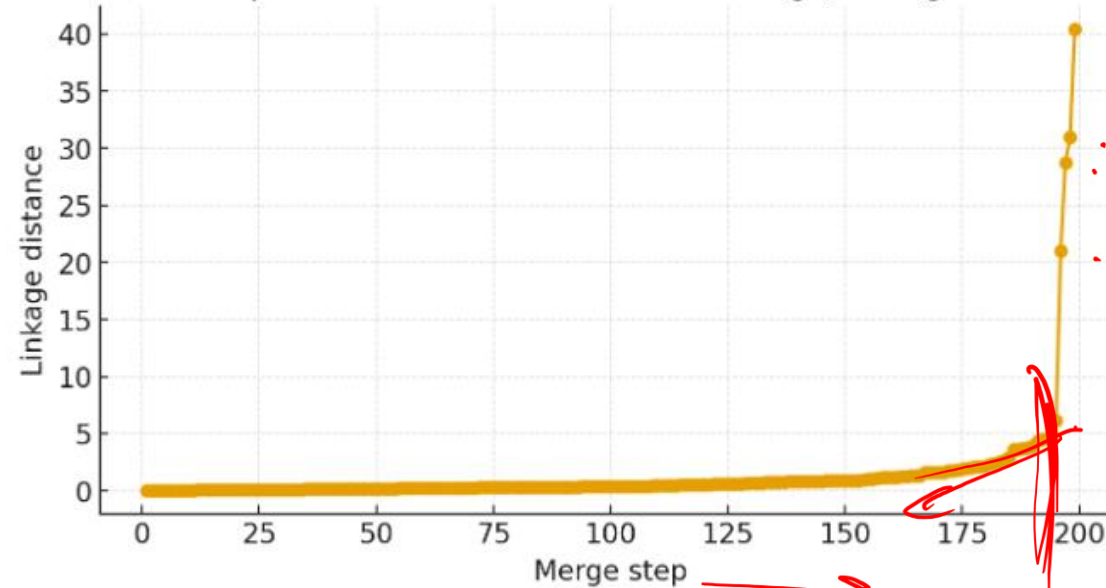
We want intra-cluster datapoints to be as close as possible to each other and inter-clusters to be as far as possible from each other

Elbow method in hierarchical clustering

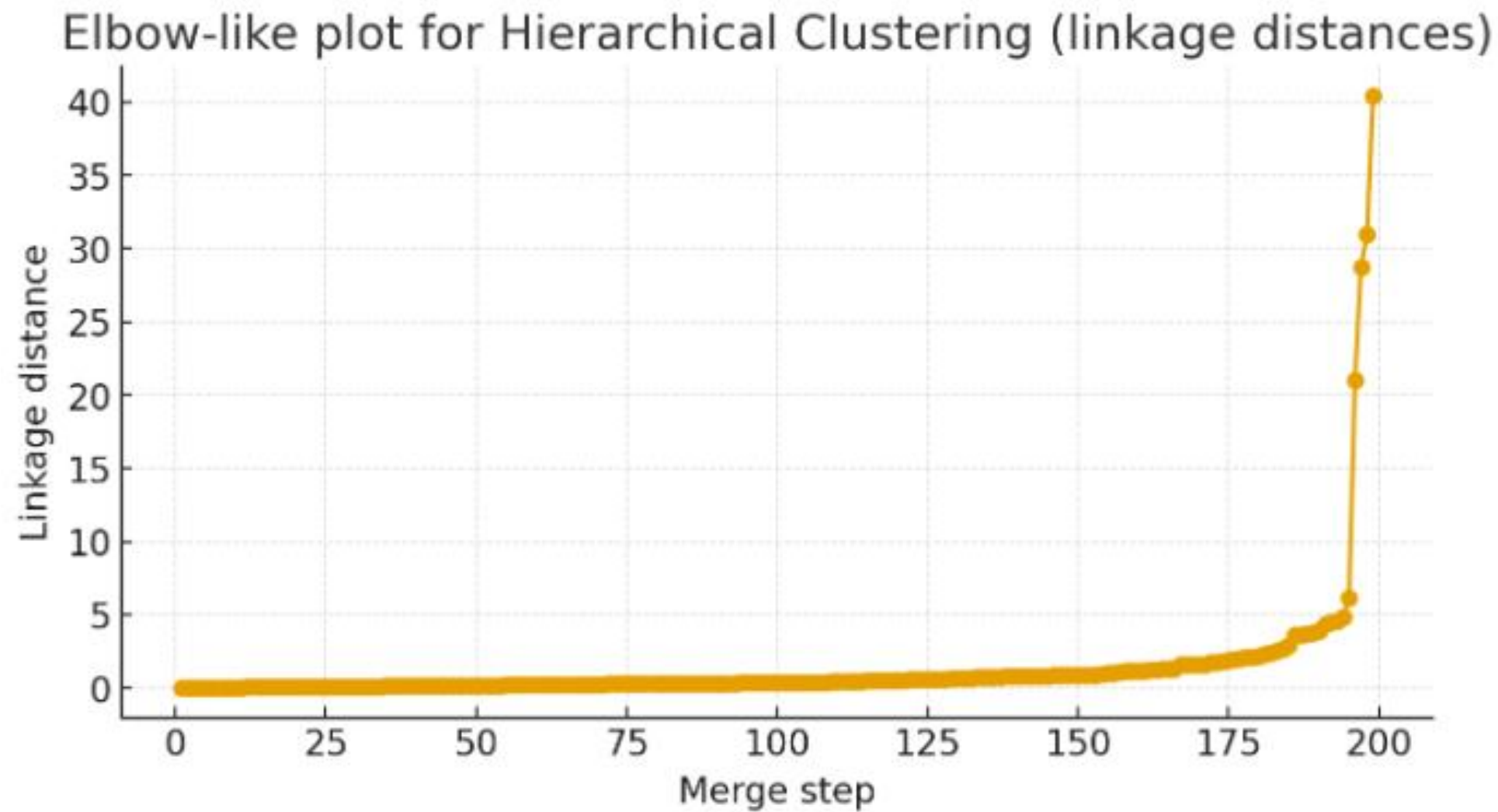
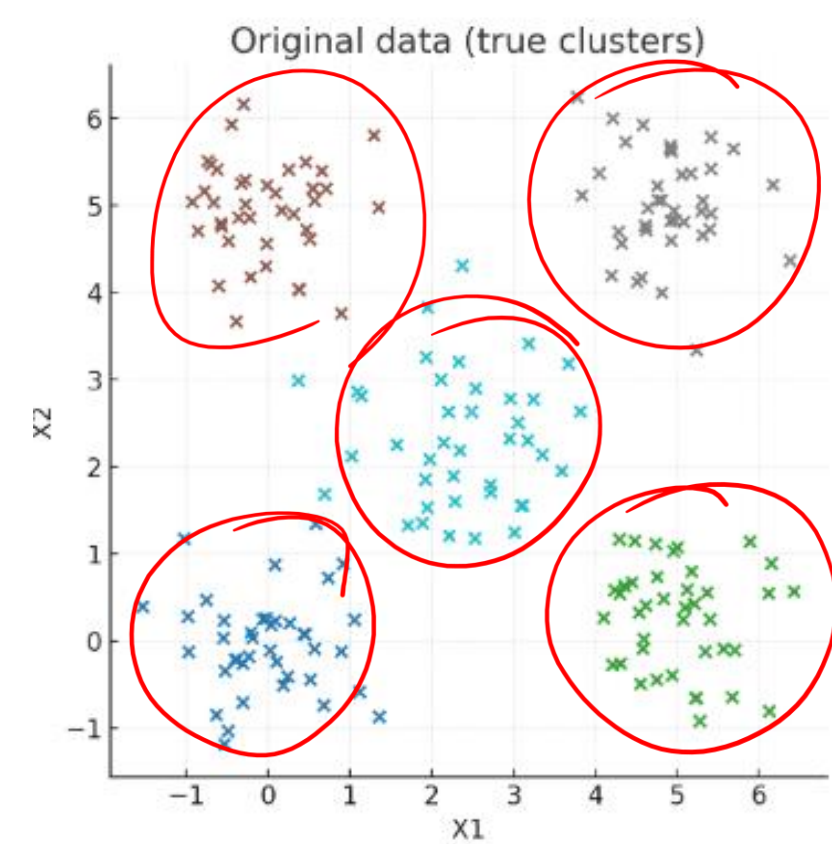
Based on sudden increases in linkage distance



Elbow-like plot for Hierarchical Clustering (linkage distances)



Elbow method in hierarchical clustering



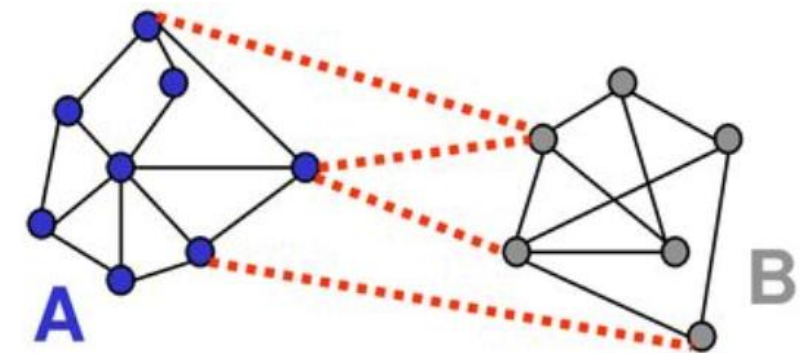
C_1 (1000) C_2 (100,000) $WC_2 > WC_1$
 Normalize

n datapoints $\rightarrow nC_2$ edges $\rightarrow \frac{n(n-1)}{2}$

The Beta-CV Measure

- Let W be the pair-wise distance matrix for all the given points. For any two point sets S and R , we define:

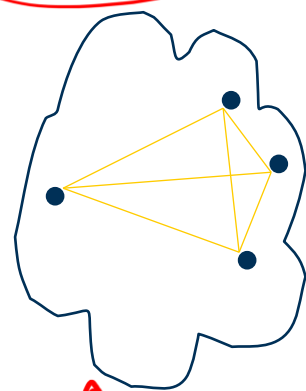
$$W(S, R) = \sum_{\mathbf{x}_i \in S} \sum_{\mathbf{x}_j \in R} w_{ij}$$



The sum of all the intracluster and intercluster weights are given as

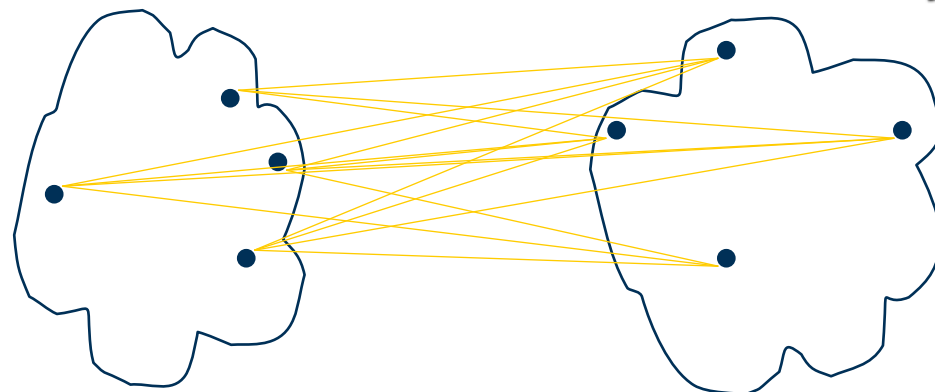
$$W_{in} = \frac{1}{2} \sum_{i=1}^k W(C_i, C_i)$$

The distance of each point is measured two times



cohesion weights

$$W_{out} = \frac{1}{2} \sum_{i=1}^k W(C_i, \bar{C}_i) = \sum_{i=1}^{k-1} \sum_{j>i} W(C_i, C_j)$$



separation weights

The Beta-CV Measure

The number of distinct intracluster and intercluster edges is given as:

$$N_{in} = \sum_{i=1}^k \binom{n_i}{2}$$

$$N_{out} = \sum_{i=1}^{k-1} \sum_{j=i+1}^k n_i \cdot n_j$$

BetaCV Measure: The BetaCV measure is the ratio of the mean intracluster distance to the mean intercluster distance:

Small

$$BetaCV = \frac{W_{in}/N_{in}}{W_{out}/N_{out}} = \frac{N_{out}}{N_{in}} \cdot \frac{W_{in}}{W_{out}} = \frac{N_{out}}{N_{in}} \frac{\sum_{i=1}^k W(C_i, C_i)}{\sum_{i=1}^k W(C_i, \bar{C}_i)}$$

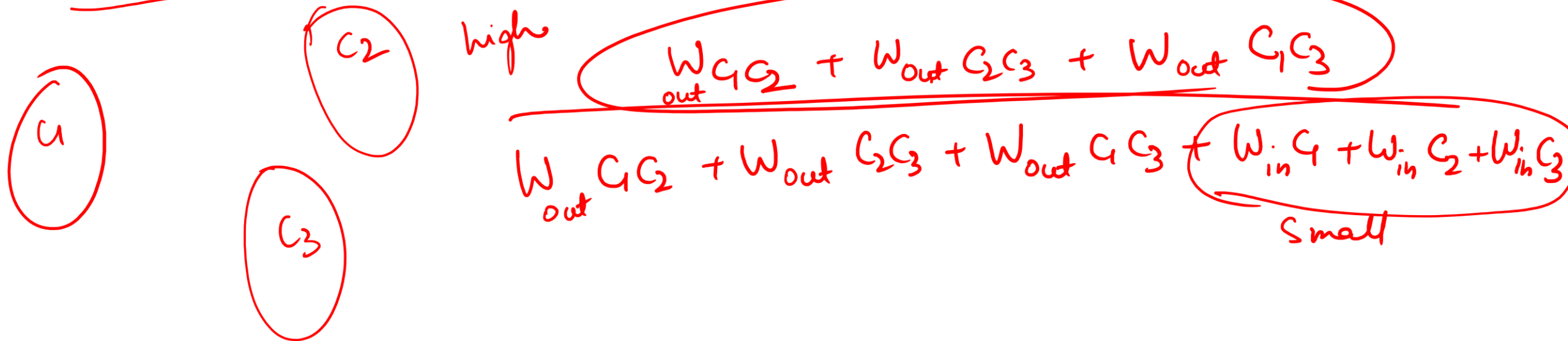
The smaller the BetaCV ratio, the better the clustering.

Normalized Cut

Normalized cut:
$$NC = \sum_{i=1}^k \frac{W(C_i, \bar{C}_i)}{\text{vol}(C_i)} = \sum_{i=1}^k \frac{W(C_i, \bar{C}_i)}{W(C_i, V)} = \sum_{i=1}^k \frac{W(C_i, \bar{C}_i)}{W(C_i, C_i) + W(C_i, \bar{C}_i)} = \sum_{i=1}^k \frac{1}{\frac{W(C_i, C_i)}{W(C_i, \bar{C}_i)} + 1}$$

where $\text{vol}(C_i) = W(C_i, V)$ is the volume of cluster C_i

The higher normalized cut value, the better the clustering

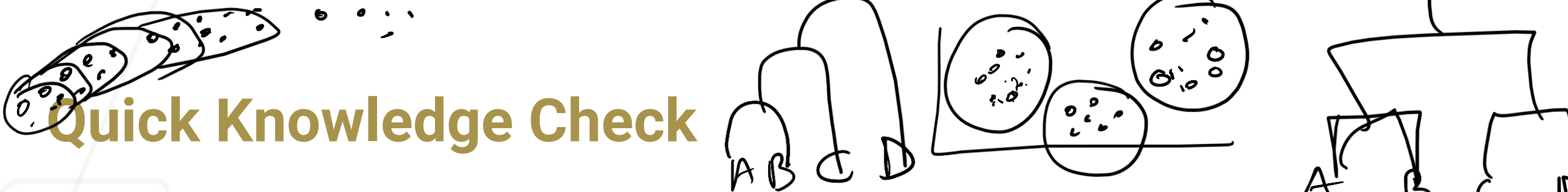


$W(C_i, C_i)$

$W(C_i, \bar{C}_i)$

Intra-cluster distance

Inter-cluster distance



Quick Knowledge Check

1. In agglomerative hierarchical clustering, do we start with: a) All data points in one cluster. b) Each data point as its own cluster
2. Which linkage method is most sensitive to chaining and outliers? a) Single link. b) Complete link. c) Average link
3. Cutting a dendrogram at a higher distance threshold produces: a) More clusters. b) Fewer clusters
4. Which of the following is a real-world use case of hierarchical clustering? a) Organizing scientific papers into a taxonomy. b) Predicting stock prices. c) Training a deep reinforcement learning agent
5. The elbow method in hierarchical clustering is based on: a) Within-cluster sum of squared errors. b) Sudden increases in linkage distance. c) Average silhouette score